

TPS System Administration Guide
for Windows XP/2003

TP06X-400

TPS R400

10/2011

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About This Document

This document provides procedures and guidelines for performing TPS system and TPS node administrative activities.

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References

The following list identifies sources of reference for material discussed in this publication.

Document Title	Doc ID
<i>System Management Configuration Guide</i>	EX08
<i>Experion Backup and Restore User's Guide</i>	BR20
<i>TPS System Implementation Guide (for Windows 2003/XP)</i>	TP08X-400
<i>TPS System Planning Guide (for Windows 2003/XP)</i>	TP10X-400

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol Definitions

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1. Introduction to TPS System Administration

1.1 Overview

Who should use this document

The *TPS System Administration Guide (for Windows 2003/XP)* provides guidelines for TPS system and TPS node administrative activities. This document contains procedures and guidelines that take the TPS system administrator through the related tasks.

Additional references

Other documents are available in support of tasks such as to plan and configure/upgrade your TPS system components. Those documents are listed below.

Document Number	Document	Related Tasks
TP10X-400	<i>TPS System Planning Guide</i>	Planning for a new or expanded TPS system.
TP08X-400	<i>TPS System Implementation Guide</i>	New implementation or upgrade of a TPS system and TPS nodes.
TPS SCN	<i>TPS Software Change Notice</i>	Last minute changes and/or workarounds when upgrading to a new release of software, or configuring a new node.
BR20	<i>Experion/TPS Backup and Restore User's Guide</i>	Includes backups and restores of TPS nodes.

1 Introduction to TPS System Administration

1.1 Overview

Prerequisite skills

This guide assumes the following PC administrative skills:

- Experience in administering Windows 2003/XP and installing Windows 2003/XP options and software.
- Experience with TPS system and TPN administration.
- Understand basic TPS system concepts, the difference between TPS system and TPN, and TPS node types.

System prerequisites and configuration notes

Please ensure the following general guidelines are true and understood before continuing with any administrative tasks on the TPS system.

- The TPS system with which you are currently working has been configured following the appropriate *TPS System Implementation Guide* procedures and is fully operational.
- The default Honeywell configuration and system policies are used as defined in this document. The administrator of the TPS system can choose to configure the system using different policies. While this is allowed, it is not the intent of this document to cover all possible configurations of policies.
- The policies and procedures defined here are to be used on a per node basis within a single Windows 2003 domain. This does not preclude the use of zero administration techniques, trusted domains, or physically separate resource and account domains, but those techniques are not described here.

2. Backup and Recovery Guidelines

2.1 Documentation for Backup and Recovery

Experion Backup and Restore User's Guide



REFERENCE - INTERNAL

Refer to the *Experion Backup and Restore User's Guide* for complete information on backup and recovery of TPS nodes.



ATTENTION

The information that was previously published in this section to support earlier TPS releases has been superseded by the *Experion Backup and Restore User's Guide*.

2 Backup and Recovery Guidelines

2.1 Documentation for Backup and Recovery

3. TPS Node Initialization Guidelines

3.1 Approach to Initialize a TPS 400 Node

Overview of node initialization process

This section describes how to load a hard drive with the Microsoft Windows 2003/XP operating system and the system hardware drivers at Honeywell factory defaults. The procedure in this section uses the EXPPlus DVD to load the default operating system image initially placed on the system at the Honeywell factory.

After performing the initialization procedure, you must use the TPS System Software CD to add all TPS software (including TPS Base) that is required for the particular TPS node type.



ATTENTION

The procedures in this section apply to TPS nodes, not to integrated Experion-TPS nodes (ES-T, ESVT, and ACE-T).

For initialization of integrated Experion-TPS nodes, refer to the *Integrated Experion-TPS Nodes User's Guide*.

To determine if system initialization is required

Normally, you initialize a node only if the hard drive becomes corrupted to such an extent that it is no longer bootable.

If a system hard drive becomes irrecoverably corrupted and the system can no longer be booted, it may be necessary to reformat the hard drive and reload the system from distribution media to reinitialize the system.

To reinitialize a system, it must be booted into MS-DOS mode. Normally this can be accomplished with the Honeywell EXPPlus DVD.

No requirement for boot diskette

For TPS R400 and greater, the EXPPlus DVD supports TPS node initialization, so creation of a boot diskette is not required.

3 TPS Node Initialization Guidelines

3.2 Node Initialization Requirements

3.2 Node Initialization Requirements

Initialization media

EXPPlus DVD Installs the Windows 2003 Server or Windows XP Professional operating system and hardware drivers with Honeywell factory defaults.

The TPS R400 Software Change Notice (SCN) lists the part number of the EXPPlus DVD that applies to the TPS release.



ATTENTION

The EXPPlus DVD does not install TPS software, such as TPS Base or the Honeywell security model. After node initialization, you must manually install this software with the **TPS System Software CD**.

Supported platforms and operating systems

The following table lists operating systems and platforms supported for TPS 400 nodes.

Operating System Supported	Node Types Supported	Platforms Supported
Windows XP Professional (for workstations)	• GUS • USR (Universal Station Replacement) • AMR (Application Module Replacement)	• WS360-based single processor Honeywell workstation • WS370-based single processor Honeywell workstation • WS470-based single/dual processor Honeywell workstation • WS490-based single/dual processor Honeywell workstation
Windows 2003 Server (Standard Edition)	• APP • AMR (Application Module Replacement) member server	• PE1600SC-based Honeywell server • PE1800-based Honeywell server • PE2850-based Honeywell server • PESC1430-based Honeywell server • PE2900-based Honeywell server • PE2950-based Honeywell server

3.3 Node Initialization Procedure

Before you begin

The node initialization procedure loads the system hard drive with the Microsoft Windows 2003/XP operating system with Honeywell factory defaults.

CAUTION

The initialization procedure reformats the entire hard drive. **All data currently on the hard drive will be lost.** Before proceeding, save any needed application and configuration data.

For more information on backing up node data, refer to the *Experion Backup and Restore User's Guide*. The document includes TPN and TPS backup procedures.



ATTENTION

The initialization procedure applies only to TPS nodes, not to integrated Experion-TPS nodes, such as ES-T, ESVT, and ACE-T.

To install operating system

Step	Action
1	Insert the EXPPlus Initialization DVD and restart the node. Note: The EXPPlus DVD replaces the TPS Initialization CD that was used on previous TPS releases.
2	On restart, when prompted, press any key to boot from CD. RESULT: The node boots from the EXPPlus DVD.
3	In Step 1 of 8 , select TPS and then click Generate .

3 TPS Node Initialization Guidelines

3.3 Node Initialization Procedure

Step	Action
4	In the Configuration Tool that launches, select the appropriate options/configurations in each tab. The defaults may be appropriate in most cases, but you may need to specify the following: - Computer Name - Company Name and other details - Windows Product Key - Disk Partition Information For Windows 2003 Server, the following information needs to be specified: - Server License Details - Server Services
5	Click Generate Configuration File on the Summary tab. Select No in response to the Write to Floppy? prompt.
6	Click Continue on Step 1 of 8.
7	Confirm to wipe out the hard disk and start the installation process. CAUTION The process formats the hard drive before starting the installation. Previous data stored on the drive will no longer be accessible after confirming the dialog.
8	In Step 4 of 8, when a dialog box appears that requests a Windows 2003 Server or Windows XP CD, insert the appropriate CD. RESULT: Windows 2003 Server or Windows XP installation begins and the node reboots.
9	Accept the EULA agreement when prompted.
10	Specify an administrator password when prompted.
11	The node restarts upon completion of initialization.
12	The node is ready for installation of TPS software from the TPS System Software CD.  ATTENTION After initialization completes, you must manually install the required TPS software to implement the TPS node.

To install factory-default TPS software

Step	Action
1	Insert the TPS System Software CD .
2	Click Install Packages from the dialog.
3	When the Welcome dialog displays, read the details and if you agree, click Next .
4	Accept the software license agreement terms by clicking Next .
5	Note the third party software information, and click Next .
6	If this dialog has been used before, the information will be filled in; if not, enter the Name, Company, License No., and Authorization No. information, and then click Next. RESULT: After the License No. and Authorization No. have been validated, the "Package Selection" dialog displays a list of the available licensed packages.
7	For GUS nodes: Install GUS Base Software. For APP nodes: Install APP Base Software.
8	In the Installable Packages dialog, retain the default selection and click OK .
9	Restart the node.
10	The node is now at the Honeywell factory-default state. After completing this initialization procedure, you can perform <u>either</u> one of the following tasks to complete the TPS node implementation: - Perform the steps in the "New TPS node implementation checklist" in the <i>TPS System Implementation Guide</i>. - Perform the "Network Install of TPS Software" procedure in the <i>TPS System Implementation Guide</i>. (You must have completed the Administrative Installation as described in this guide.)  REFERENCE - INTERNAL For detailed instructions, refer to the <i>TPS System Implementation Guide</i>.

3 TPS Node Initialization Guidelines

3.3 Node Initialization Procedure

4. Windows 2003 Domain Controller Guidelines

4.1 FSMO Roles in a TPS System

Overview of FSMO roles

This section provides instructions to set up Flexible Single Master Operation (FSMO) roles and FSMO role owners in a TPS system. Although Windows 2003 supports peer domain controllers that share information dynamically through multi-master replication, a few functions may need to remain on a single domain controller exclusively. These functions are known as Flexible Single Master Operation (FSMO) roles.

TPS system administrators should become familiar with FSMO roles. The following roles emulate legacy Primary Domain Controller (PDC) functionality when running in mixed mode:

- PDC Master (Does not exist in Native Mode)
- RID (Relative Identifier) Master (Does not exist in Native Mode)
- Infrastructure Master

Why FSMO roles are important

A failure of FSMO roles in a mixed mode Windows 2003 domain will result in reduced functionality. In particular, the peer domain controllers in Windows 2003 are unable to create new accounts, but are still able to perform the critical authentication duties.

Where to assign FSMO roles

In general, in a mixed mode environment, the administrator assigns FSMO roles to the first Windows 2003 domain controller in the Windows 2003 domain. In the event of a failure, the roles may need to be transferred to an alternate domain controller. This process is known as transferring or seizing an FSMO role.

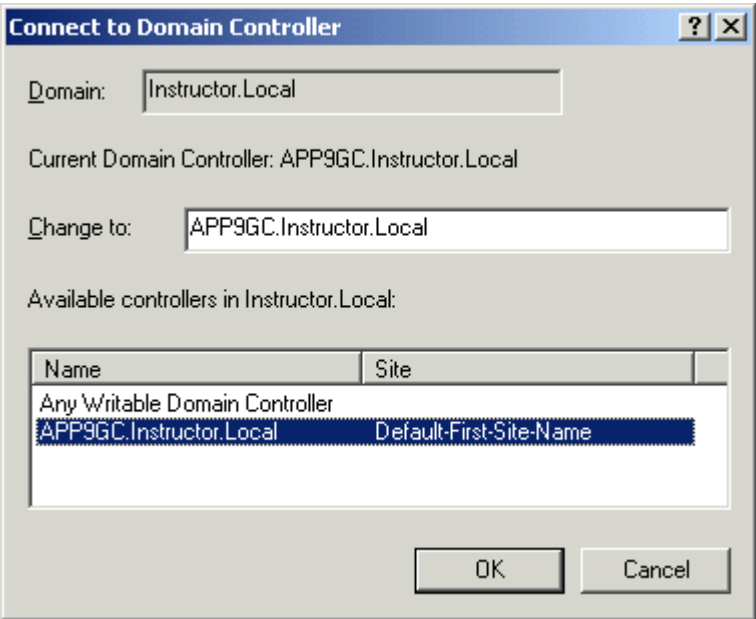
Seizing a role is necessary only when the existing role master is permanently inoperable. The concept of transferring or seizing a role is very similar to that of promoting a BDC to a PDC in Windows NT 4.0. Essentially, Windows NT 4.0 transferred the PDC role to a BDC during the promotion process.



ATTENTION

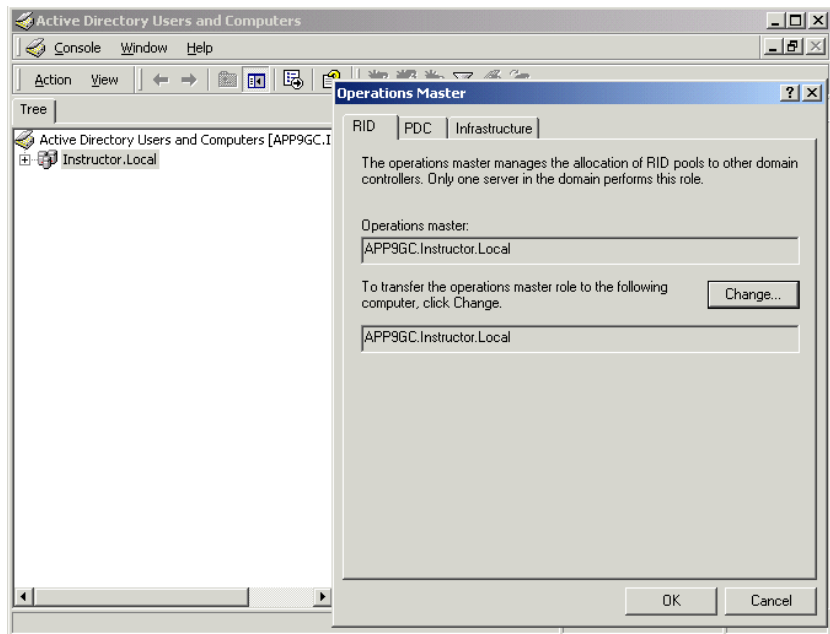
If a domain controller that hosts the FSMO roles will be out of service for any length of time, you should transfer the roles to an alternate domain controller.

To view and change the FSMO role owners

Step	Action
1	Open the Active Directory Users and Computers Snap-In on a domain controller as follows. Start > Programs > Administrative Tools > Active Directory Users and Computers
2	a) Right-click the domain name in the scope pane and select Connect to Domain Controller. b) Select the domain controller that you would like to become the new owner of the FSMO roles. c) Click OK. 

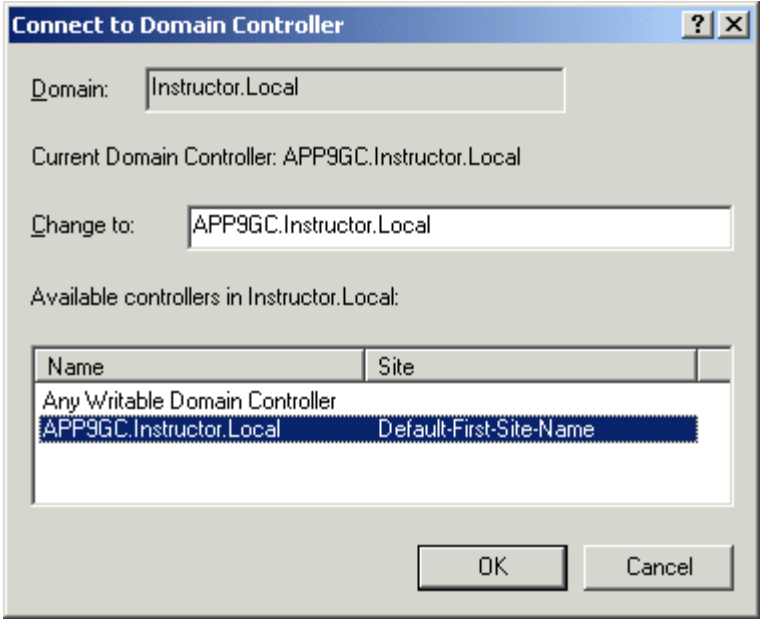
4 Windows 2003 Domain Controller Guidelines

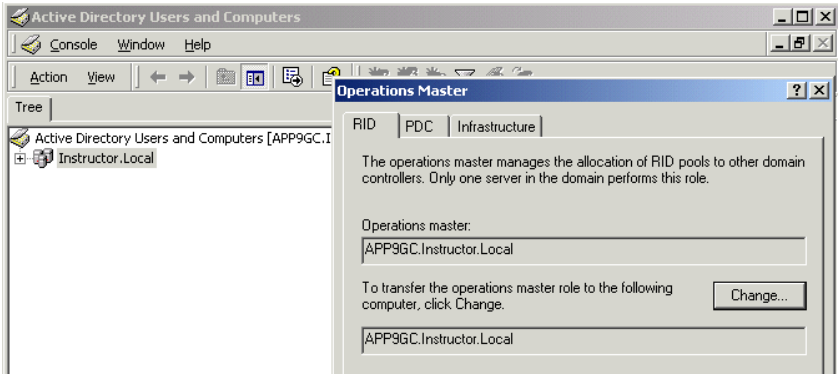
4.1 FSMO Roles in a TPS System

Step	Action
3	Right-click the domain name in the scope pane and this time select Operations Masters from the menu.
	 The screenshot shows the 'Active Directory Users and Computers' console window. The 'Tree' pane on the left shows the domain 'Instructor.Local'. The 'Operations Master' dialog box is open, displaying the 'RID' tab. The dialog box contains the following text: 'The operations master manages the allocation of RID pools to other domain controllers. Only one server in the domain performs this role.' Below this, it says 'Operations master:' followed by a text box containing 'APP9GC.Instructor.Local'. There is a 'Change...' button next to it. At the bottom, there is an 'OK' button and a 'Cancel' button.
4	Each tab represents a role master and displays the current location of the current role master. Validate the Operations Master for each tab (RID, PDC, and Infrastructure), and click OK if they all are correct.

To transfer FSMO role owners

If the node that hosts FSMO roles fails and it is evident that recovery will be extended or impossible, it is necessary to transfer the FSMO ownership. In addition, if you plan to take down the node hosting the FSMO role owners for an extended time, you should transfer the roles before taking the node offline.

Step	Action
1	Open the Active Directory Users and Computers Snap-In on a domain controller. Start > Programs > Administrative Tools > Active Directory Users and Computers
2	Right-click the domain name in the scope pane and select Connect to Domain Controller.... Then Select the domain controller that you would like to become the new owner of the FSMO roles. Click OK 

Step	Action
3	Again, right-click the domain name in the scope pane. This time select Operations Masters from the menu.  RESULT: Each tab represents a role master and displays the current location of the current role master.
4	Select the tab of the role you wish to transfer (typically, that is all tabs).
5	Verify that the server name in the lower edit box is the name of the domain controller that is to become the new owner of the FSMO roles.
6	Click Change , and then click OK .

To seize role ownership

If the existing Operation Master node is offline, it may be necessary to seize the Role Ownership. Seizing roles is possible only through the Windows command line utility ntdsutil.exe. Ntdsutil.exe is outside the scope of this document.



REFERENCE - EXTERNAL

Refer to Microsoft documentation for assistance to use the command line utility Ntdsutil.exe to seize roles.

4 Windows 2003 Domain Controller Guidelines

4.1 FSMO Roles in a TPS System

5. TPS Security

5.1 Overview of TPS Security

Purpose of TPS security

The purpose of TPS security is to provide every user with the minimum rights and the functionality required to perform required tasks effectively. In a Plant Control Network (PCN), the philosophy is to deliberately set it up as a "closed" network, such that the various PCN nodes have sufficient functionality to meet the needs of the overall PCN. In general, however, the PCN nodes will not include the wider ranging rights and functionality required to meet the needs of a general Plant Information Network (PIN).

By adopting this approach, it is relatively straightforward to make the PCN secure. When additional rights and functionality are required to meet the more general needs of the PIN and its users, adding them becomes a conscious, deliberate decision. Details of the required changes can be designed based on the precise, application-specific needs of the users. This approach is consistent with normal networking practices.

Most of the following sections in this document discuss security as it applies to a domain environment. If you need operator lockdown in a workgroup environment, one option is the Honeywell-supplied Local Computer Policy.

For instructions to apply a Local Computer Policy, see section 5.14, "Enabling the Local Security Policy in a Workgroup or Standalone Environment."

Honeywell high security model

The Honeywell high security model follows the Microsoft High Security Model guidelines which list a series of steps recommended to secure Windows 2003/XP nodes.

The configuration philosophy of the Honeywell high security model is enforced through the combination of Group Policy Objects, Domain user accounts, Domain Groups, Local user accounts, Local Groups, Proxy Files, Computer and User Policies, and scripts that reside at both the domain controller and workstation systems.

A TPS system depends on the existence of specific Local users, Domain users, and Domain (Global) Groups. This document describes each of these areas.



ATTENTION

Honeywell highly recommends that you apply this file system security model to protect against unauthorized file access. This is especially important if you plan to use SafeView or any application that employs a browse window.

5 TPS Security

5.1 Overview of TPS Security

Honeywell high security model installation packages

The default TPS security model is applied on Windows XP Professional and Windows 2003 Server (non-Domain Controllers) through manual installation of high security software packages. Honeywell supplies the high security model as two separately installable packages on the **TPS System Software CD**:

- **Experion PKS – High Security Domain Controller Package** – This package creates the security components on the Domain Controller, including the secure Group Policy Objects (GPOs) and global groups.
- **Experion PKS – High Security Workstation Package** – This package creates the security components on workstation or member server nodes. The user installs this security package on each workstation and member server node in the domain to set up the local policies, accounts, and users.

Domain Controller - Experion PKS – High Security Domain Controller Package

- Creates Group Policy Objects (GPOs) for each user type,
- Creates global groups used as filters on the GPOs,
- Creates global accounts, and
- Assigns the global accounts to the global groups.

Experion PKS – High Security Workstation Package

- Creates local groups and users,
- Creates proxy files for managing Honeywell application access,
- Ensures the workstation DCOM settings support interoperation within the system,
- Establishes local computer policy settings, and
- Provides the **linkdomaingroups.vbs** script that you must execute to make specific domain groups members of local groups – this integrates the workstation into the domain security model.

5.2 Domain Controllers in a TPS System

Domain Controller purpose in the TPS system

The TPS system makes extensive use of Windows 2003 domain architecture. Although some components are configurable outside of a domain, the domain architecture is important when bringing individual components together to form a cohesive system.

A Windows domain brings many advantages, one of which is simplification of security administration.

TPS domain configuration provides the underpinnings for TPS System Management. TPS System Management requires the existence of a Windows domain.

More advanced domain models and security configurations are possible. For details, see section Customizing the TPS Security Model, "Customizing the TPS Security Model."

Domain Controller roles in Windows 2003

Windows 2003 provides only one type of domain controller, and each domain controller is considered a peer of other domain controllers within the same domain.

The concepts of Primary Domain Controller (PDC) and Backup Domain Controller (BDC) no longer exist in name.

5.3 Group Policies in a TPS System

TPS system – Group Policy descriptions

The following table shows the Honeywell high security model Group Policies and their descriptions. The table lists the Group Policy Objects (GPOs) that the **Experion PKS – High Security Domain Controller** package creates in the Active Directory, and each GPO's associated Global Group that "filters" the scope of the group object.

Group Policy Name	Filter (Global Group)	Description
Administrator Group Policy	Administrators	A minimally restricted user environment. This account is typically used for day-to-day Windows XP/2003 and DCS administrative tasks.
Engineer Group Policy	Engineers	A restricted user environment that allows members to perform relevant process control activities. Administrative actions in the Windows XP/2003 environment are limited.
Operator Group Policy	Operators	A very restricted user environment that permits members of this group to run only allowed applications. Typically, members of this group have a specified logon script that automatically starts relevant applications. Usage of the Microsoft Internet Explorer browser is limited to intranet or local applications, such as Knowledge Builder.
Supervisor Group Policy	Supervisors	Similar to Operators in terms of severely limited Windows capabilities.
View Only Group Policy	View Only Users	A very restricted user environment that permits members of this group to run only allowed applications with only the ability to view.
ACK View Only Group Policy	View Only Users	A very restricted user environment that permits members of this group to run only allowed applications with only the ability to view. Members also have privileges to acknowledge alarms.



ATTENTION

Honeywell recommends that users not modify the TPS policies directly. Instead, copy the TPS policy to an alternate name and modify the new policy.

How to view Group Policies

The contents of each Group Policy can be examined by selecting the Active Directory Users and Computers Snap-In in the Programs>Administrative Tools program group. A comprehensive list of configuration settings is not included in this document.

Perform the following steps to examine the Group Policy settings.



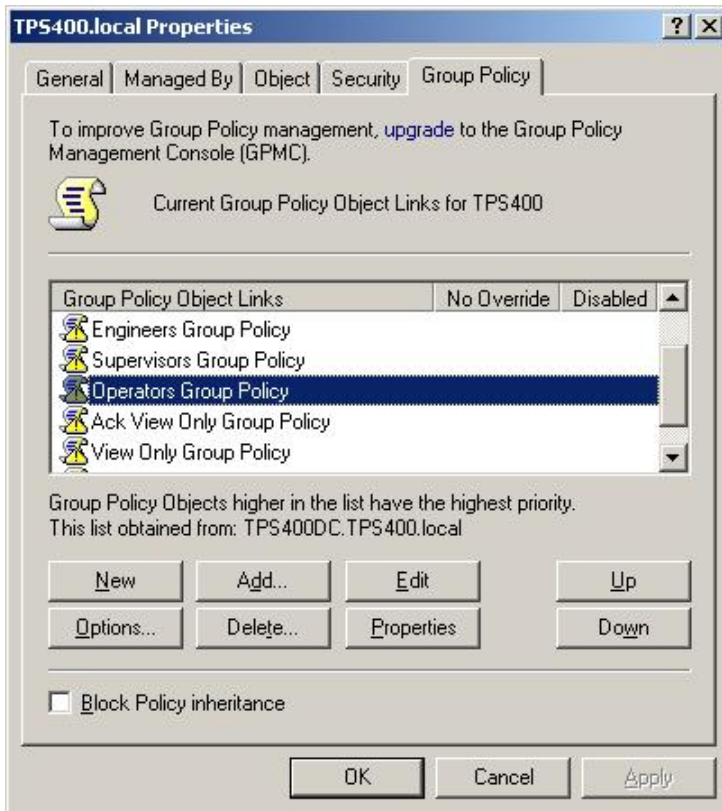
REFERENCE - EXTERNAL

Refer to the Microsoft documentation for additional details on viewing and configuring Group Policy Objects.

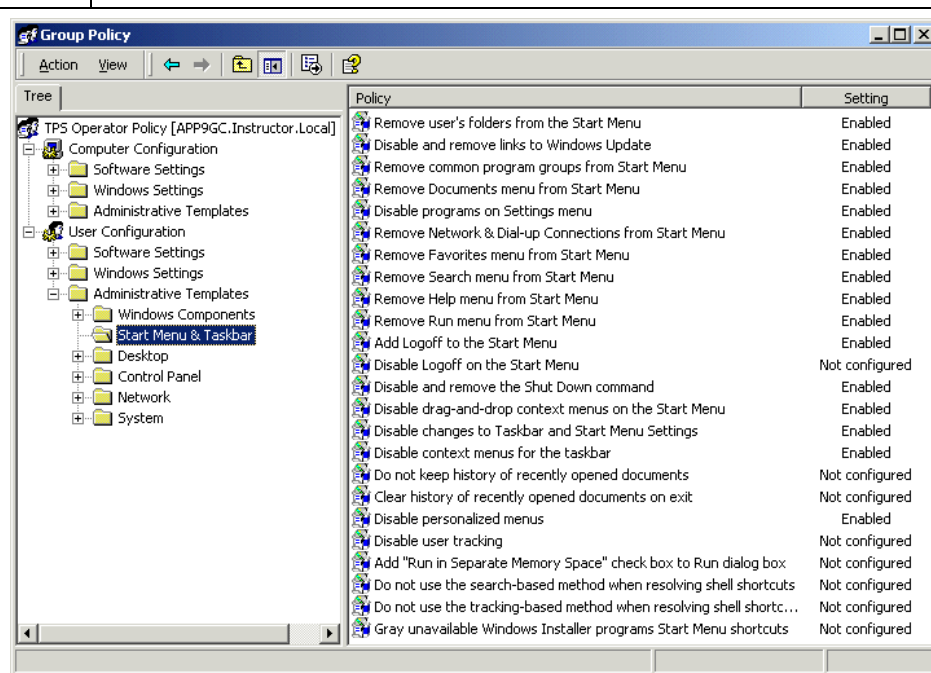
Step	Action
1	Open the Active Directory Users and Computers Microsoft Maintenance Console (MMC) Snap-In on a domain controller. Note: The Group Policies can also be viewed from a Windows XP Professional or Windows 2003 Server node by opening the Group Policy Snap-In. Start > Programs > Administrative Tools > Active Directory Users and Computers
2	Right-click the domain name and select Properties .
3	Select the Group Policy tab in the dialog box.

5 TPS Security

5.3 Group Policies in a TPS System

Step	Action
4	Examine Group Policy settings by selecting the applicable Group Policy Name and clicking Edit. 

Step	Action
	Policies can be viewed or modified by browsing the items within the scope pane. Most TPS-configured settings are applied within the User Configuration settings. The following screen shows settings for the TPS Operator group policy. Settings for configuring the Start Menu are shown below.

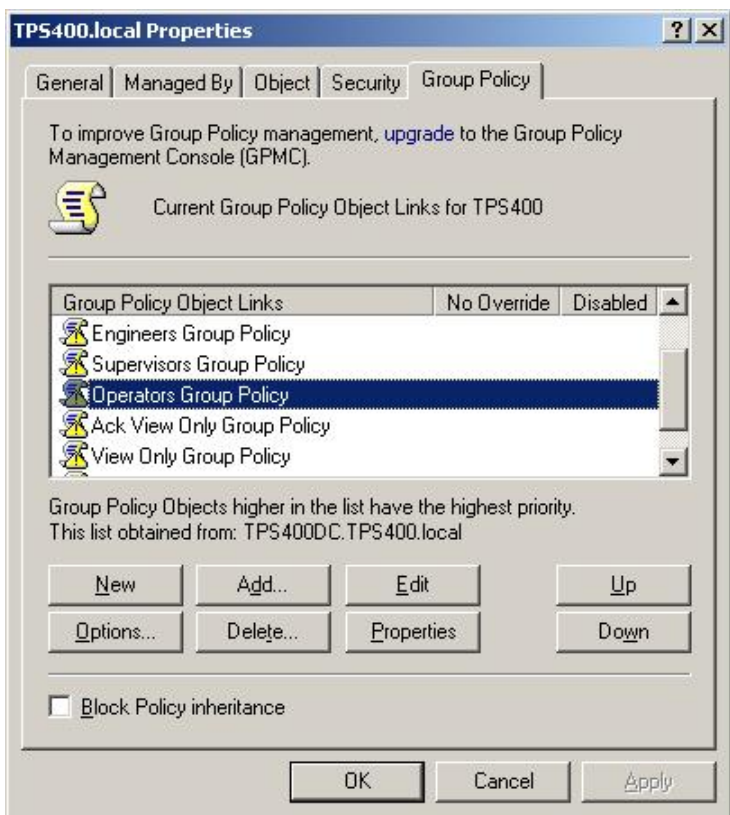


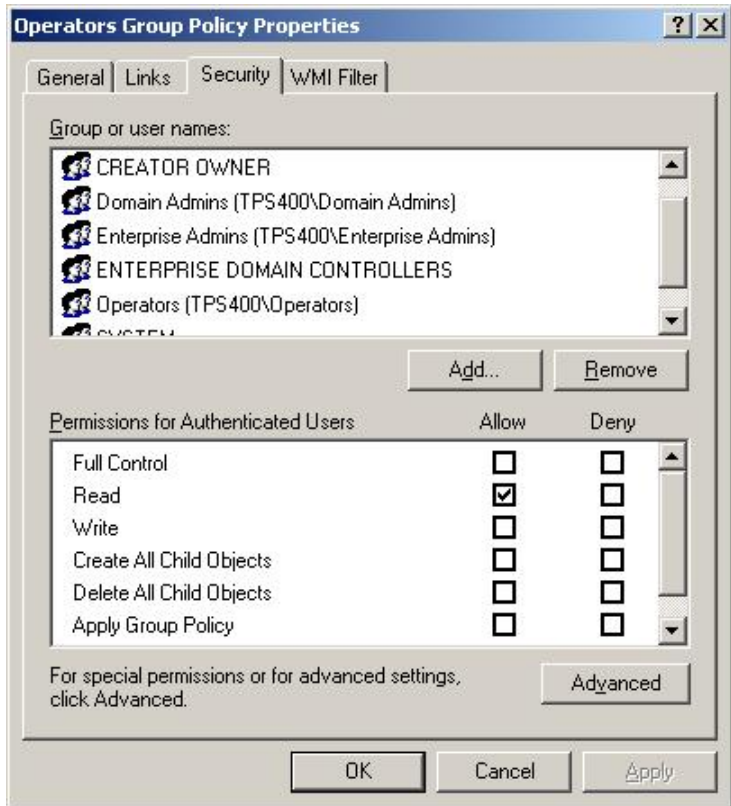
ATTENTION

Honeywell recommends that users not modify the TPS policies directly. Instead, copy the TPS policy to an alternate name and modify the new policy.

5 TPS Security

5.3 Group Policies in a TPS System

Step	Action
5	Ensure the appropriate security filter is applied. Group Policies apply only to users or groups that fall within the selected security filter. To ensure the appropriate security filter is applied, select the applicable Group Policy name, and click the Properties.
	

Step	Action																					
	Example Results: The following graphic shows the TPS Operator Group Policy security settings that include the Apply Group Policy access control entry for the Operators global group. This restricts the TPS Operator Group Policy to only members of the Operators group.  The screenshot shows the 'Operators Group Policy Properties' dialog box with the 'Security' tab selected. The 'Group or user names' list contains several entries, including 'Operators (TPS400\Operators)'. Below this list are 'Add...' and 'Remove' buttons. The 'Permissions for Authenticated Users' section contains a table with columns 'Allow' and 'Deny'. The 'Apply Group Policy' row has the 'Allow' checkbox checked. At the bottom, there is an 'Advanced' button and 'OK', 'Cancel', and 'Apply' buttons. <table><tr><th>Permissions for Authenticated Users</th><th>Allow</th><th>Deny</th></tr><tr><td>Full Control</td><td>☐</td><td>☐</td></tr><tr><td>Read</td><td>☒</td><td>☐</td></tr><tr><td>Write</td><td>☐</td><td>☐</td></tr><tr><td>Create All Child Objects</td><td>☐</td><td>☐</td></tr><tr><td>Delete All Child Objects</td><td>☐</td><td>☐</td></tr><tr><td>Apply Group Policy</td><td>☒</td><td>☐</td></tr></table>	Permissions for Authenticated Users	Allow	Deny	Full Control	☐	☐	Read	☒	☐	Write	☐	☐	Create All Child Objects	☐	☐	Delete All Child Objects	☐	☐	Apply Group Policy	☒	☐
Permissions for Authenticated Users	Allow	Deny																				
Full Control	☐	☐																				
Read	☒	☐																				
Write	☐	☐																				
Create All Child Objects	☐	☐																				
Delete All Child Objects	☐	☐																				
Apply Group Policy	☒	☐																				

5 TPS Security

5.4 TPS Users and Groups



ATTENTION

Pay particular attention to who is assigned the Apply Group Policy security access control entry. Inadvertently selecting the Apply Group Policy entry for Authenticated Users, Everyone, Administrators, and/or Domain Admins groups can have extremely undesirable results, resulting in a loss of administrative capabilities.

For example, if the Operator Group Policy had both the Operators and the Authenticated Users groups selected to receive Apply Group Policy, then ALL authenticated users (including administrators) would receive the restricted settings contained within the Operator Group Policy.

Microsoft recommends that a defensive group policy technique be used - select the DENY Apply Group Policy checkbox for administrative accounts to ensure they never inadvertently receive group policies.

5.4 TPS Users and Groups

TPS specific users and groups

To simplify administrative tasks, the Experion PKS - High Security Policy – Domain Controller installation package creates TPS system-specific global (domain) users and global (domain) groups on the Domain Controller. In addition, group membership for both global and local groups is established, as shown in the following table.

TPS global groups

The TPS system makes extensive use of global (domain) and local groups to apply security in an extensible manner. Global groups are created and maintained within the Windows 2003 Active Directory. These groups have a scope of a Windows 2003 domain. Global groups are used as a mechanism to organize domain user accounts.

The site's system administrator should create user accounts and add such accounts into the appropriate groups, as shown in the following table.

Global (Domain) Group Name	Interactive Users?	TPN Access Level*	Purpose
DCS Administrators	Yes	N/A	Separates the role of the DCS administrator from the IT administrator. Members of this Global group do not have permission to perform Domain Controller configuration. They do, however, have the right to install and configure software on TPS nodes. This is possible because DCS Administrators are members of the local administrators group.
DCS Domain Servers	No	N/A	Local Servers encapsulate users who are assigned as the distinguished principal for DCOM servers and/or Windows 2003 services.
Intimate Users	No	1	Has the highest access level to the underlying system. Has access to all default proxy files.
Point Builders	No	2	Supports applications that may require a higher security access level.
Continuous Controls	NO	3	Supports applications that may require a higher security access level. Members equate to the TPN Continuous Control function level.
Programs	No	4	For CL application users.
Engineers	Yes	5	For DCS control or system engineers.
Supervisors	Yes	6	Typically for shift leaders located in the control room. This group is severely restricted, but has the permissions to • Log on to a workstation, and • Shutdown and restart the workstation.
Operators	Yes	7	For control room operators. Typically, members have very restricted permissions on TPS nodes.
View Only Users	Yes	8	Members have view-only access to the process.

* The TPN Access Level indicates a rank of security access level on the TPN (LCN), which range from the highest access (Intimate Users - 1) to the lowest access (View Only – 8).

5 TPS Security

5.4 TPS Users and Groups

Local Groups

All local node security is applied using either local Windows 2003/XP groups or local TPS groups. You should be aware of the following:

- global groups permit only global users as members, and
- local groups permit global users and local users, in addition to global groups as members.

For more information, refer section Customizing the TPS Security Model, "Customizing the TPS Security Model."

For TPS GUS and APP nodes, the user executes the following script during node implementation:

\Program files\Honeywell\WkStaSecurity\linkdomaingroups.vbs



ATTENTION

The script that joins the local workstation security to the domain security (linkdomaingroups.vbs) should be executed after a domain is established and the node is a domain member.

Failure to do so will render distributed communications inoperable due to lack of appropriate security credentials.

Local group names and Global group associations

The following table provides details on the global (domain) groups that are copied to local groups on each TPS node by the execution of the script linkdomaingroups.vbs.

This Global (Domain) Group...	Is added to this Local Group on the TPS node...
DCS Administrators	Administrators, Honeywell Administrators
Ack View Only Users	Local Ack View Only Users
Engineers	Local Engineers, Engineering Repository Users
Intimate Users	Local Intimate Users
Operators	Local Operators
Point Builders	Local Point Builders
Programs	Local Programs
DCS Domain Servers	Local Servers
Supervisors	Local Supervisors
View Only Users	Local View Only Users
Continuous Controls	TPS Continuous Controls

5 TPS Security

5.4 TPS Users and Groups

Domain server groups

The following table lists the groups created on a Domain Controller and their details (where Domain Group means the same as Global Group).

Groups created on Domain Controller					
Group	Type...	Created by... **	Members...	Type...	Created by... **
Domain Users	Domain	OS	Administrator	Domain User *	OS
			TPSComServer	Domain User *	TPS
			TPSLocalServer	Local User *	Admin
			TPSApp	Domain User *	TPS
Administrators	Local	OS	Administrator	Domain User	OS
			Domain Admins	Domain Group	OS
			DCS Administrators	Domain Group	TPS
Backup Operators	Local	OS	Domain Admins	Domain User	Admin
			DCS Administrators	Domain Group	TPS
Replicator	Local	OS			
Administrators	Domain	TPS	TPSAdministrator	Domain User *	Admin
Continuous Controls	Domain	TPS	(None)		
Engineers	Domain	TPS	Engineer	Domain User *	Admin
Intimate Users	Domain	TPS	(None)		
Operators	Domain	TPS	Operator	Domain User *	Admin
Point Builders	Domain	TPS	(None)		
Programs	Domain	TPS	TPSApp	Domain User *	TPS
Supervisors	Domain	TPS	Supervisor	Domain User *	Admin
View Only Users	Domain	TPS	View Only	Domain User *	Admin
Administrators	Local	TPS	DCS Administrators	Domain Group	TPS

Groups created on Domain Controller					
Group	Type...	Created by... **	Members...	Type...	Created by... **
Continuous Controls	Local	TPS	Continuous Controls	Domain Group	TPS
Engineers	Local	TPS	Engineers	Domain Group	TPS
Intimate Users	Local	TPS	Intimate Users	Domain Group	TPS
Operators	Local	TPS	Operators	Domain Group	TPS
Point Builders	Local	TPS	Point Builders	Domain Group	TPS
Programs	Local	TPS	Programs	Domain Group	TPS
Servers	Local	TPS	DCS Domain Servers	Domain Group	TPS
Supervisors	Local	TPS	Supervisors	Domain Group	TPS
View Only Users	Local	TPS	View Only Users	Domain Group	TPS

* Indicates user's primary group

** Indicates how the group was created:

"Created By"	Meaning
Admin	Created by Administrator
OS	Created by Windows 2003/XP operating system
TPS	Created by Experion PKS – High Security Domain Controller package

5 TPS Security

5.4 TPS Users and Groups

User account types

The following table provides information for different types of user accounts.

User Name	Type	Interactive User?	Purpose
TPSApp	Global	No	This user account is the account under which CL Applications run. It is configured to allow only logging on as a batch job. Note: The TPSApp account (used by the CLServer on APP nodes) has no equivalent in the Experion environment. In TPS R400, the CLServer installation package creates this global account (if it does not already exist) as a member of the Programs group.
TPSComServer	Global	No	This user account is typically used for HCI COM/OPC Servers provided as part of the TPS system. This account is configured to log on only as a batch job.
TPSLocalServer	Local	No	Local servers that are configured to run on a TPS node use this account.

Creating additional user accounts

After executing the security model package, create user accounts as appropriate. To simplify administration, each new user should be added to a global group. For example, adding an operator level account involves creation of the user account using the users and groups MMC snap-in. A new user is now able to log on to TPS nodes. In addition, a new user is subject to all the applicable group policy restrictions.



REFERENCE - EXTERNAL

Refer to Microsoft documentation for details on adding operator level accounts in this manner.

TPS group configuration guidelines

Users with multiple group memberships

If a user is member of two or more TPS groups, then the group policy having the lowest access privileges will override the others. For example, if a user is member of the 'DCS Administrators' and 'Engineers' global (domain) groups, then the 'Engineer' policy will override the 'DCS Administrators' policy - the 'Engineer' policy prohibits its members from accessing the registry.

User access to the registry



ATTENTION

Registry access is disabled whenever a domain user is a member of any of the following TPS global (domain) groups:

- Engineers
 - Supervisors
 - Operators
 - View only users
-

By default, the Domain Administrator has access to the registry; however, if you add the Domain Administrator user to the 'Engineers' global (domain) group (to gain Native Window 'Engineer' access privilege), then the Domain Administrator will be prohibited from accessing the registry.

If access to the registry is needed, you can Disable the **Prevent Access to Registry Editing Tools** setting in the group policy.

'Engineer' privileges

In TPS400, the original local group 'TPS Local Administrators' is replaced by the local group 'Honeywell Administrators'. Members of 'TPS Local Administrators' previously got 'Engineer' privilege by default – in TPS 400, members of 'Honeywell Administrators' do NOT get 'Engineer' privilege.

In TPS R400, upon execution of the script linkdomaingroups.vbs, the global group 'DCS Administrators' becomes a member of local groups 'Administrators' and 'Honeywell Administrators' – therefore, members of the 'DCS Administrators' group do not get 'Engineer' privileges.

5.5 Applying TPS Security Model

Overview

You apply the TPS security model on Windows XP Professional and Windows 2003 Server (non-Domain Controllers) when you install the **Experion PKS – High Security Workstation Package** from the TPS System Software CD. The security package can be reinstalled; however, that should not be necessary unless security issues arise. To further simplify security implementation, TPS makes extensive use of local groups. You should recognize that

- global (domain) groups permit only global (domain) users as members, and
- local groups permit global (domain) users, local users, and global (domain) groups as members.

All local TPS node security uses either local Windows groups or local TPS groups – this promotes extensible security and supports various domain models. It also allows Honeywell to apply TPS security to files, registry keys, and user rights when the machine loads from the TPS System Software CD. **Note:** If security were applied using global TPS groups, it would have the undesirable effect of requiring a domain controller before the TPS security scripts could be used.

Run LINKDOMAINGROUPS to assign global groups to TPS local groups

After you establish the Windows 2003 domain, you must install the **Experion PKS – High Security Domain Controller Package** on the domain controller – this installation creates specific Group Policies and Global Groups in the Active Directory.

Then, you must add the TPS node to the domain and execute the following script on the TPS node – this script assigns specific global (domain) groups to TPS local groups:

\Program files\Honeywell\ WkStaSecurity\linkdomaingroups.vbs



ATTENTION

Execute the script LINKDOMAINGROUPS.VBS *after* you establish a domain and the TPS node is a domain member. This script joins the local security implementation with the domain model.

Failure to do so will render distributed communications inoperable due to lack of appropriate security credentials.

For additional considerations concerning user group membership, refer to section 5.4, "TPS group configuration guidelines".

Details for TPS local groups on each TPS node

The following table lists local groups that are on a TPS node, and lists the local users, global users, and global groups that are typical members of the local groups.

This Local Group on the TPS node...	Is created by...*	And contains these members...	Of this type...	Created by...*
Administrators	OS	Administrator	Domain User	OS
		Domain Admins	Domain Group	OS
		DCS Administrators	Domain Group	TPS
Backup Operators	OS			
Replicator	OS			
Users	OS	Global Users	Local User	TPS
		TPSLocalServer	Local User	TPS
Honeywell Administrators	TPS	Administrators	Domain Group	TPS
Local Ack View Only Users	TPS	Ack View Only Users	Domain Group	TPS
Local Continuous Controls	TPS	Continuous Controls	Domain Group	TPS
Local Engineers	TPS	Engineers	Domain Group	TPS
Local Intimate Users	TPS	Intimate Users	Domain Group	TPS
Local Operators	TPS	Operators	Domain Group	TPS
Local Point Builders	TPS	Point Builders	Domain Group	TPS
Local Programs	TPS	Programs	Domain Group	TPS
Local Servers	TPS	DCS Domain Servers	Domain Group	TPS
		TPSLocalServer	Local User	TPS
Local Supervisors	TPS	Supervisors	Domain Group	TPS
Local View Only Users	TPS	View Only Users	Domain Group	TPS
	ATTENTION In TPS R400, user accounts that are members of the global group "DCS Administrators" and the local groups "Administrators" and "Honeywell Administrators" <u>do not</u> have Engineer privilege.			
* OS	These are created by Windows 2003/XP operating system			
* TPS	These Local Groups are created on the TPS node by the Experion PKS – High Security Workstation Package. These Global Users and Global Groups are created on the domain controller by the Experion PKS-High Security-Domain Controller Package, and are added as members of Local Groups on the TPS Node by the script linkdomaingroups.vbs.			

5.6 Password Security and Synchronization

Recommendation for password security



ATTENTION

Experion and TPS nodes ship with default passwords. To avoid having passwords compromised, Honeywell highly recommends that you change the defaults to strong passwords (such as to include one or more spaces).

Password tool

The Honeywell password tool displays Windows services and DCOM servers that are registered to run under a selected account. After you enter secure passwords for specific Honeywell accounts, use the password tool to synchronize the Windows and DCOM services associated with those accounts.

Use this tool to synchronize Logon Accounts and Passwords on Windows Services and DCOM servers. This tool does not change the Accounts' password, use after changing Accounts' password.

1. Select an Account Name from the drop down list
2. Enter Password for the Account or Enter a New User Name and Password
3. Press Apply New Account

Account Name: ID-SV18\TPSLocalServer

New Credentials

User Name: ID-SV18\TPSLocalServer

Password: [masked]

Apply New Account

Associated Servers

Server Name	Internal Name	Server Type
Hci.spsdss - System Performance Server - exe Server	{0336C76F-6...	COM
HCI_TPNServer - exe Server	{ADF6AEBB-...	COM

Cancel

When to use the password tool

Use the Honeywell password tool after changing passwords for any of the following accounts:

Domain accounts:

TPSComServer

DCSComServer

TPSApp

Note: The CLServer installation package creates the TPSApp global account (if it does not already exist) as a member of the Programs group.

Local accounts:

LocalComServer

TPSLocalServer

Mngr



ATTENTION

The password tool needs to be run on every node that is using the changed account.

Installation of the password tool

The password tool is installed by the **System Management Runtime** package from the Honeywell **TPS System Software CD**.

5 TPS Security

5.6 Password Security and Synchronization

How to use the password tool

After entering secure passwords for specific accounts, perform the following steps to synchronize the Windows and DCOM services associated with those accounts.

Step	Action
1	To call up the password utility, select one of the following, depending on the node type: Start>Programs > Honeywell Experion PKS>System Management > Windows Services & DCOM Servers Log on tool. OR Start>Programs > Honeywell TPS System Management. > Windows Services & DCOM Servers Log on tool.
2	Select an Account Name from the dropdown list. RESULT: Under Associated Servers, the tool lists the Windows services and DCOM servers associated with the selected account name.
3	Type-in the password for the account.
4	Select Apply New Account . RESULT: If the password you just entered matches the account's password, then the tool synchronizes the associated Windows services and DCOM servers with the new account information. A message appears indicating that the operation succeeded.
5	Repeat steps 2-4, as necessary for each changed account.
6	Run the password tool and repeat this procedure on every node that is using the changed account(s).



ATTENTION

The password tool does not change the password on the account.

5.7 Diagnosing Security-Related Issues

MMC Security Configuration and Analysis

The Microsoft Security Configuration and Analysis tool is a valuable resource for debugging security related issues. The following procedure can be used to compare the currently applied security against the predefined TPS security settings.



ATTENTION

In some cases, differences between the installed settings and TPS settings are perfectly acceptable. These cases are usually caused by software application installations that apply incremental security beyond what is described in the TPS security templates. For example, the logon rights for a batch job and the service user log on rights frequently do not match the initial TPS settings.

Procedure to run Security Configuration and Analysis

Follow these steps to run and examine the Security Configuration and Analysis snap-in.

Step	Action
1	Start the Microsoft Management Console as an administrator. At the command line or in the Run a program dialog, type mmc .
2	Add the Security Configuration and Analysis Snap-in, as follows. a. Select Console\Add Remove Snap-In . b. Select Security Configuration and Analysis from the snap-in list c. Click Add. d. Click Close.
3	Open the applicable TPS Security Database File. Right-click the Security Configuration and Analysis item in the scope pane of the MMC display, and Select Open Database.
4	For Windows XP Professional and Windows 2003 Server (Non-Domain Controllers), navigate to and then select \\windows\\security\\database\\wkssdb.sdb .

5 TPS Security

5.7 Diagnosing Security-Related Issues

Step	Action
5	To perform the analysis, Right-click the Security Configuration and Analysis item in the scope pane of the MMC display, and then select Analyze Computer Now in the menu.
6	View the security in the result pane.

Items shown as a green check mark indicate a successful comparison.

Items shown as a red x indicate a mismatch between the TPS defaults and the actual security settings on the computer.

You should examine red items by double-clicking to identify the discrepancy. Some items may have additional security settings that were applied by individual software application installations.

7	Optionally, view the text-based security log: Select the Security Configuration and Analysis item in the scope pane to display the security log. This log file is located in the local \windows\security\logs directory.
---	--

Reapply TPS security settings for security mismatches

If security mismatches are discovered, the TPS security settings can be reapplied to the system by right-clicking the **Security Configuration and Analysis** item in the scope pane and then selecting the **Configure Computer Now** target.



REFERENCE - EXTERNAL

Refer to Microsoft Security Configuration and Analysis references in the Windows online help for additional information.

5.8 Customizing the TPS Security Model

Purpose of TPS security model

The default TPS security model (installed through the Experion EPKS High Security workstation and domain controller packages) applies various security settings to provide a secure computing environment for Windows 2003/XP-based process control nodes. The default TPS security model (also referred to as the Honeywell high security model) is extensible—it is up to the customer to determine the appropriate modifications to meet site-specific security needs. The following sections provide an overview of some optional extensions to the TPS security model.

Domain model

Domain models can be one of the following types:

- Single Master domain
- Console-Based Domain

5.9 Implementing Single Master Domain Architecture

What a single Master domain represents

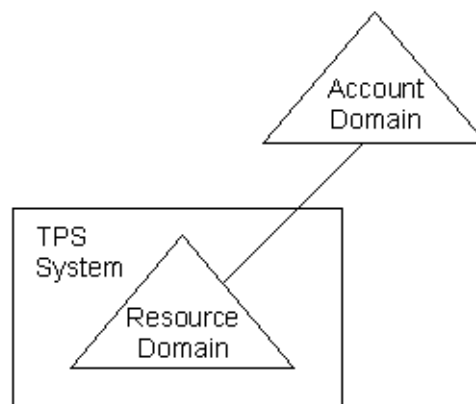
A Single Master Domain is also known as an Account/Resource domain. This very common domain structure maintains most of the user accounts in a single Account domain. Resource domains (such as process control domains) are implemented using either one-way or two-way trust relationships with its parent. The benefit is that centralized administration of users across an enterprise is achieved.

Typically, supervisors and engineers have logon accounts in the Account domain for their desktop machines. The default TPS security model requires that these users create new accounts in the process control domain. This results in significant duplication of effort. In addition, it becomes a management burden for the individual users to keep track of passwords (which can change at different intervals) on multiple domains. The Single Master domain implementation will eliminate this duplication of effort by storing all user accounts in the Account domain.

Example of a process control domain

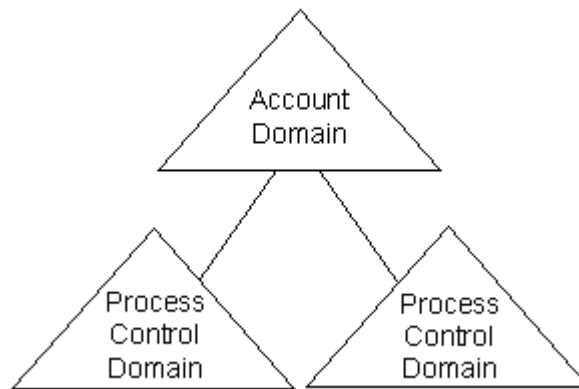
The Resource domain in the following diagram represents a process control domain. The diagram shows common, single, master domain architecture. Desktop user accounts are managed within the corporate Account domain. Process control domains are one or more Resource domains maintaining a one way trust relationship with the Account domain. Often a router is implemented between IT and Process Control LANs for security and bandwidth isolation purposes.

The following diagram represents a domain controller for the Account domain residing on the same LAN as the process control equipment to improve authentication reliability.



Example of multiple process control domains

The following diagram expands upon the diagram above by showing multiple process control domains leveraging a single Account domain. In this case, Account domain users can use their desktop accounts to log on to multiple process control domains throughout the enterprise.

**Integrating process control and IT domains**

Customers interested in achieving integration between Process Control and IT Domains should perform the standard TPS installation steps, and then perform the steps in the following "Implementing a single master domain" procedure. This procedure permits use of Account domain user accounts in order to log on to local process control nodes.

When logging on to process control nodes using an Account domain user, the policies and logon scripts from the Account domain will take effect. This can produce undesirable results, because the goals of desktop user management and process control user management are different. Plan carefully before you perform the following procedure.

5 TPS Security

5.9 Implementing Single Master Domain Architecture

Implementing a single master domain

Follow the general steps that follow to implement a single master domain.

Step	Action	Details
1	Add Global Groups.	It may be beneficial to create additional global groups in the IT Domain that correspond with the process control related groups, such as Operators or Engineers.
2	Add Users to Groups.	Add the IT user accounts to the newly created global groups.
3	Add the IT Global Groups to the Process Control Local Groups.	On each TPS node, add each IT global group to the equivalent or appropriate local group. Refer to the table that follows for a common configuration.

Local groups that participate in single master domain

The following table lists local group memberships for TPS nodes that participate in a single master domain architecture, such as Account/Resource. In this case, IT_DOMAIN represents the Account domain and PC_DOMAIN represents the process control or Resource domain. The TPS security script LINKDOMAINGROUPS establishes the PC_DOMAIN group memberships on each node when executed. The IT_DOMAIN memberships must be added manually or through a script file created by the customer.

Local Group	Local Group Membership
Local Operators	IT_DOMAIN\Operators PC_DOMAIN\Operators
Local Engineers	IT_DOMAIN\Engineers PC_DOMAIN\Engineers
Local Supervisors	IT_DOMAIN\Supervisors PC_DOMAIN\Supervisors
DCS Administrators	IT_DOMAIN\DCS Administrators PC_DOMAIN\DCS Administrators

5.10 Implementing a Console Based Domain Architecture

Why should you use a console-based domain?

Some of your site standards may require additional security control at a more detailed level than a single Operators, Engineers, and/or Supervisors group. In the following scenario, assume you would like to organize your nodes into a console configuration (a collection of nodes represents a console). Additionally, you would like to restrict Operator access on one or more consoles.

Example scenario of a console based domain

The following diagram describes a process control domain containing six TPS nodes. The nodes are organized as follows:

- Console one contains three nodes
- Console two contains two nodes
- Console three contains a single node.

In this architectural scenario, security should allow a given operator to only log on and operate a node within a given console.

Implementing a console-based domain

To implement this architecture, first implement the Honeywell Experion PKS - High Security model. After the domain is created and each TPS node is a member, execute the following customization procedure. Details are included for our example scenario.

Step	Action	Result
1	Add Global Groups.	Add new global groups in the process control domain to represent smaller collections of nodes, such as Consoles. For our scenario: CONS1_OPS, CONS2_OPS, CONS3_OPS
2	Add Users to Global Groups.	Add the user accounts to the applicable groups. For our scenario: User joesmith is a member of the CONS1_OPS group. User janesmith needs access to nodes as an operator in multiple consoles. Add janesmith to both CONS1_OPS and CONS3_OPS. Note: If janesmith needs access to all nodes in all consoles as an operator, add janesmith to the Operators group supplied by TPS.

5 TPS Security

5.10 Implementing a Console Based Domain Architecture

Step	Action	Result
3	Add Global Groups to Local Groups.	On each TPS node, add new global groups to the equivalent or appropriate local group. Refer to the following table for a common configuration.

Results in our scenario of console based domain implementation

In the previous procedure, example entries were made based on our scenario that has the following results. For our scenario, on nodes 1-3, add the CONS1_OPS global group to the local Operators group. Users joesmith and janesmith are members of the CONS1_OPS group—thus, they have appropriate permissions to log on to node 1.

Local Group	Nodes 1-3 Local Group Membership
Local Operators	PC_DOMAIN\Operators (Automatically created by TPS) PC_DOMAIN\CONS1_OPS (Add this manually)

For nodes 4-5 add the CONS2_OPS global group to the local Operators group. Because user joesmith is NOT a member of the CONS2_OPS group or the default Operators group, he does NOT have access to nodes 4 and 5.

Local Group	Nodes 4-5 Local Group Membership
Local Operators	PC_DOMAIN\Operators (Automatically created by TPS) PC_DOMAIN\CONS2_OPS

On node 6 add the CONS3_OPS global group to the local Operators group. Because user joesmith is NOT a member of the CONS3_OPS group or the default Operators group, he does NOT have access to nodes 4 and 5. User janesmith is a member of the CONS3_OPS global group, so she has access to node 6.

Local Group	Node 6 Local Group Membership
Local Operators	PC_DOMAIN\Operators (Automatically created by TPS) PC_DOMAIN\CONS3_OPS

Customization of Group Policies

You can create additional Group Policies to customize the user desktop environment. The customization of Group Policies is beyond the scope of this document.



REFERENCE - EXTERNAL

[Refer to Group Policies in Microsoft documentation for additional details.](#)

5.11 HCI/OPC Server Capability Files

Using ACLs of HCI/OPC server capability files

The Access Control Lists (ACLs) of the HCI/OPC server capability files are used to determine who has the capability to use HCI/OPC server-secured methods.

Following is an example of the Configure Control page, used to secure the TPN server methods:

Method	Capability
OPCRead	Continuous Control
OPCWrite	Point Builder
Shutdown	Supervisor

5 TPS Security

5.11 HCI/OPC Server Capability Files

In the previous screen example, the DCS administrator secured the TPN Server using the following methods:

- The **Continuous Control** capability file is used to secure the “OPCRead” method.
- The **Point Builder** capability file is used to secure the “OPCWrite” method.
- The **Supervisor** capability file is used to secure the “Shutdown” method.

Secured method access security

The following table lists the TPS security access level and the resulting method access level for each of the eight standard capability files.

Capability File Security and Method Access Levels		
Capability Filename	Access Security Level	Method Access Level
Intimate User	Highest Access Security (Level 8)	1 – Allows Least Access
Point Builder	Access Security Level 7	2
Continuous Control	Access Security Level 6	3
Program	Access Security Level 5	4
Engineer	Access Security Level 4	5
Supervisor	Access Security Level 3	6
Operator	Access Security Level 2	7
View Only	Lowest Access Security (Level 1)	8 –Allows Most Access

Microsoft file permissions (special or advanced)

The Access Control Lists (ACLs) of the different capability files are used to build the HCI/OPC security level structure.

The capability file ACLs are designed so that the following TPS group members have "RX" (Read/Execute) access to each capability file:

- Every member of the corresponding TPS group for the capability file
- Every member of a TPS group at a higher security level

Legend of Microsoft file permissions (special or advanced)

Below is a list of letter codes used to represent the Microsoft file permissions.

- R = Read data
- D = Delete
- W = Write data
- P = Change permissions
- X = Execute
- O = Take ownership

Note: Any user with the RWDP or RWDPO permission is considered to have full access.

ACL entries for View only, Operator, Supervisor, and Engineer

Refer to the following tables for a description of the ACLs that determine the HCI/OPC security level structure.

Standard Capability File ACL Entries – Table 1 of 2				
Capability Filename (Access Security Level) →	View Only	Operator	Supervisor	Engineer
	(Lowest – 1)	(2)	(3)	(4)
Capability File Groups				
Intimate Users (TPS Domain Group)	RX	RX	RX	RX
Point Builders (TPS Domain Group)	RX	RX	RX	RX
Continuous Controls (TPS Domain Group)	RX	RX	RX	RX
Programs (TPS Domain Group)	RX	RX	RX	RX
Engineers (TPS Domain Group)	RX	RX	RX	RX
Supervisors (TPS Domain Group)	RX	RX	RX	Supervisors Not Listed
Operators (TPS Domain Group)	RX	RX	Operators Not Listed	Operators Not Listed

5 TPS Security

5.11 HCI/OPC Server Capability Files

Standard Capability File ACL Entries – Table 1 of 2				
Capability Filename (Access Security Level) →	View Only	Operator	Supervisor	Engineer
	(Lowest – 1)	(2)	(3)	(4)
View Only Users (TPS Domain Group)	RX	View Only Users Not Listed	View Only Users Not Listed	View Only Users Not Listed
Other TPS Entries				
DCS Administrators (TPS Domain Group)	RWDP	RWDP	RWDP	RWDP
DCSDomainServers (TPS Domain Group)	R	R	R	R
Windows Users/Groups				
Administrators (Windows Local Group)	RWDP	RWDP	RWDP	RWDP
Backup Operators (Windows Local Group)	RWDP	RWDP	RWDP	RWDP
Replicator (Windows Local Group)	RWDPO	RWDPO	RWDPO	RWDPO

ACL entries for Program, Continuous Control, Point Builder, and Intimate User

Standard Capability File ACL Entries – Table 2 of 2				
Capability Filename	Program	Cont. Control	Point Builder	Intimate User
(Access Security Level) →	(5)	(6)	(7)	(Highest – 8)
Capability File Groups				
Intimate Users (TPS Domain Group)	RX	RX	RX	RX
Point Builders (TPS Domain Group)	RX	RX	RX	Point Builders Not Listed

Standard Capability File ACL Entries – Table 2 of 2				
Capability Filename	Program	Cont. Control	Point Builder	Intimate User
Continuous Controls (TPS Domain Group)	RX	RX	Cont. Ctrls. Not Listed	Cont. Ctrls. Not Listed
Programs (TPS Domain Group)	RX	RX	Programs Not Listed	Programs Not Listed
Engineers (TPS Domain Group)	Engineers Not Listed	Engineers Not Listed	Engineers Not Listed	Engineers Not Listed
Supervisors (TPS Domain Group)	Supervisors Not Listed	Supervisors Not Listed	Supervisors Not Listed	Supervisors Not Listed
Operators (TPS Domain Group)	Operators Not Listed	Operators Not Listed	Operators Not Listed	Operators Not Listed
View Only Users (TPS Domain Group)	View Only Users Not Listed	View Only Users Not Listed	View Only Users Not Listed	View Only Users Not Listed
Other TPS Entries				
DCS Administrators (TPS Domain Group)	RWDP	RWDP	RWDP	RWDP
Windows 2003/XP Users/Groups				
Administrators (Windows Local Group)	RWDP	RWDP	RWDP	RWDP
Backup Operators (Windows Local Group)	RWDP	RWDP	RWDP	RWDP
Replicator (Windows Local Group)	RWDPO	RWDPO	RWDPO	RWDPO

5.12 HCI/OPC Server Capability File Customization

Why customize a standard capability file?

The standard capability files are designed to be used in environments where every user account that is created is added to one of the TPS defined groups. In this environment, every HCI or OPC application will run under an account that fits into the TPS Server method security.

If an account does not have “RX” access permission to a capability file that is securing a desired server method, then the capability file’s ACL must be edited to grant the mandatory “RX” access permission to the account or the account’s group.

Capability file – Customization example

Scenario: The HCI client runs under an account that is a member of the “PlantSeven” Windows 2003 domain (global) group.

The HCI client needs to access a TPN Server’s “Shutdown” method that is secured with the “Engineer” capability file.

The APP node in the “BoilerPlants” TPS domain hosts the TPN Server.

Solution: For this example, the following actions must be performed to customize the capability file:

- Use Windows Explore to find the “BoilerPlants.nam” file in the following directory:
\Windows\System32\Repl\Export\HWIAC\Security
- Right-click the "Engineer" capability file in the "<domain name>\Security" directory and click **Properties**.
- Select the **Security** tab and click the **Permissions** button.
- Grant RX access permission to the “PlantSeven” Windows 2003 domain group.
- Perform TPS domain replication to distribute the updated “Engineer” capability file to the TPS domain members.

For further information, refer to section Customizing HCI Server Priority Files, "Customizing HCI Server Priority Files."

Mandatory capability file ACL entries

The ACLs of all capability files **MUST** include the following table entries:

Mandatory Capability File ACL Entries	
Group or User	Permissions
Administrators (Windows Local Group)	RWDP
Backup Operators (Windows Local Group)	RWDP
Replicator (Windows Local Group)	RWDPO
DCS Administrators (TPS Domain Group)	RWDP
DCS Domain Server (TPS Domain Group)	R

**ATTENTION**

The above entries must be added to the ACLs of any capability file that is created by the TPS system administrator.

5 TPS Security

5.13 Customizing HCI Server Priority Files

5.13 Customizing HCI Server Priority Files

Purpose of HCI server priority files

The HCI server priority files that are used for TPS system security and are located on the node that hosts the HCI component, in the following directory:

`\hwiac\system\security`

HCI server priority files permissions

The following table lists the HCI server priority file security permissions as set up by the TPS Configuration Utility at the time of installation. Note that only the suffix of the HCI server priority filename is given. All of the filenames are preceded with the following:

tpn_priority

Standard Priority File ACL Entries - Table 1 of 2				
Priority Filename (Priority Level) →	Two (Lowest – 2)	Three 3	Four 4	Five 5
Administrators	RWDP	RWDP	RWDP	RWDP
Backup Operators	RWDP	RWDP	RWDP	RWDP
Continuous Controls	RX	RX	RX	RX
Engineers	RX	RX	RX	RX
Intimate Users	RX	RX	RX	RX
Operators	RX	RX	RX	RX
Point Builders	RX	RX	RX	RX
Programs	RX	RX	RX	RX
Replicator	RWDPO	RWDPO	RWDPO	RWDPO
Supervisors	RX	RX	RX	RX
DCS Administrators	RWDP	RWDP	RWDP	RWDP
View Only Users	RX	-	-	-

Standard Priority File ACL Entries - Table 2 of 2					
Priority Filename (Priority Level)	six	Seven	eight	nine	ten
	6	7	8	9	(Highest - 10)
Administrators	RWDP	RWDP	RWDP	RWDP	RWDP
Backup Operators	RWDP	RWDP	RWDP	RWDP	RWDP
Continuous Controls	RX	RX	RX	RX	RX
Engineers	RX	RX	RX	RX	RX
Intimate Users	RX	RX	RX	RX	RX
Operators	RX	RX	RX	RX	RX
Point Builders	RX	RX	RX	RX	RX
Programs	RX	RX	RX	RX	RX
Replicator	RWDPO	RWDPO	RWDPO	RWDPO	RWDPO
Supervisors	RX	RX	RX	RX	RX
DCS Administrators	RWDP	RWDP	RWDP	RWDP	RWDP
View Only Users	-	-	-	-	-

5.14 Enabling the Local Security Policy in a Workgroup or Standalone Environment

To implement the High Security Policy in a workgroup environment or for a standalone node, you need to enable the local security policy on each computer.

Note: If you use domains, you do not need to perform this procedure.

The Lockdownlocalusers.bat file enables the local security policy and changes the access security permissions to the \system32\GroupPolicy directory.

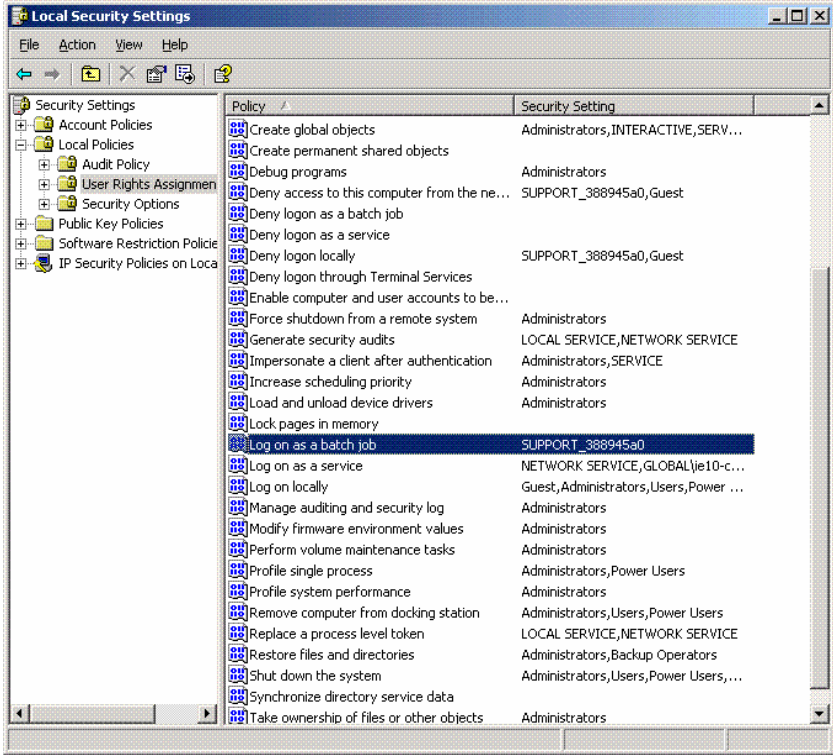
The security permissions are changed such that the local Administrator account does not have access to the GroupPolicy directory; consequently, the policy is not applied when the Administrator logs on to the computer.

The following steps will be necessary to enable the Administrator account to perform administrative tasks on the computer.

Step	Action
1	Open a command prompt.
2	Change the directory to c:\Program Files\Honeywell\WkstaSecurity .
3	Type the command, lockdownlocalusers.bat .

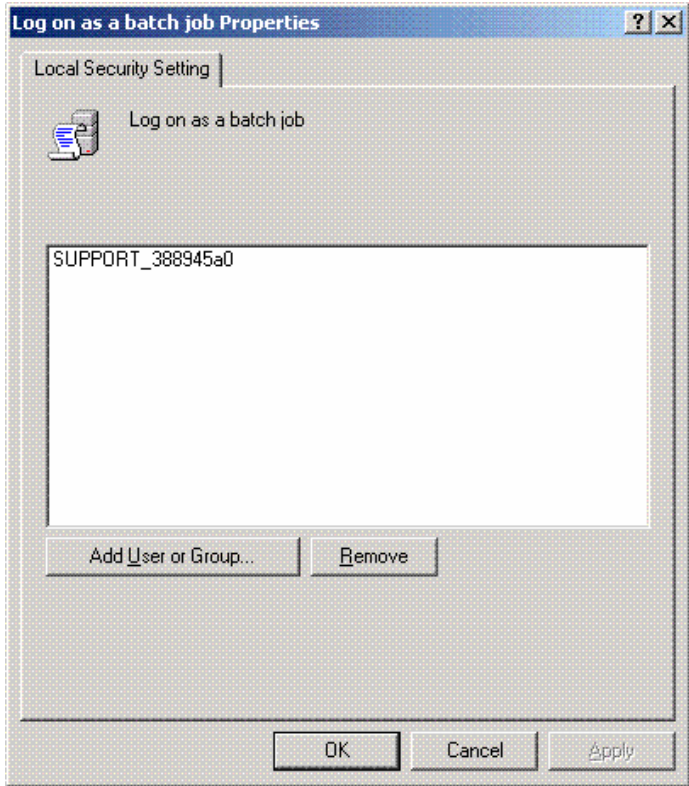
5.15 Viewing the Local Security Settings

Use the following steps to view the local security settings and edit their policy assignments:

Step	Action
1	Click Start > Settings > Control Panel . The Control Panel window appears.
2	Double-click Administrative Tools . The Administrative window appears.
3	Double-click the Local Security Policy . The Local Security Settings window appears.
4	In the Left pane, from the tree view select the Security Settings > Local Policies > User Rights Assignment.  The Policy and its Security Setting details are listed in the right pane.

5 TPS Security

5.15 Viewing the Local Security Settings

Step	Action
	TIP For example, in the preceding figure, the right to Log on as a batch job is limited only to SUPPORT_388945a0.
5	Double-click a policy from the policy column (for example, Log on as a batch job) to edit the assignment. The Properties dialog box for the selected Policy appears: 
To add a User or Group	
6	Click Add User or Group . The Select Users or Groups dialog box appears.
7	Type the user or group you want to add in the Enter the object name to select box.
8	Click OK . The user or group you have selected is added to that policy.

Step	Action
9	Click OK to close the Properties dialog box.
To remove a User or Group	
10	Select the user or group you want to remove.
11	Click Remove . It removes the policy assigned to it.
12	Click OK to close the Properties dialog box.

5.16 Localizing TPS Security Configuration

Overview

The TPS 400 system leverages the Microsoft Security Configuration Editor and security settings are compatible with all localized versions of Windows 2003. However, TPS scripts, which establish group membership based on Windows 2003 built-in groups, will require a minimal localization effort.

Use the following steps as a guideline to localize the TPS security configuration for Windows XP Professional and/or Windows 2003 Server (non-domain controller).

To localize workstations and servers (Non-Domain Controller)

Use the following steps as a guideline to localizing your workstations and servers on each node in a non-domain controller setting.

Step	Action
1	Copy the LINKDOMAINGROUPS .VBS file to a new file name (iLINKDOMAINGROUPS): Copy this application: \program files\ HoneywellWkStaSecurity\ LINKDOMAINGROUPS.VBS To this location: \program files\HoneywellWkStaSecurity\ iLINKDOMAINGROUPS.VBS .
2	Open the iLINKDOMAINGROUPS .VBS security command script.
3	Replace the following Windows 2003 English versions of built-in users and groups with the localized equivalents: • Administrators • Replicator
4	From the command line, execute the iLINKDOMAINGROUPS. VBS command script. RESULT: Security will be reapplied to the system.

6. Recommendations for Managing a TPS System

6.1 Domain Controller Recommendations

Domain Controller platform recommendations

The minimum requirements for a Windows 2003 Domain Controller are:

- 300 MHz or faster processor
- Windows 2003 Server (standard edition)
- 128 MB RAM
- 3 GB of available disk space (for data files and Active Directory storage)

For maximum system reliability, establish a Domain Controller on a server grade PC with a RAID 5 controller and three hot-swappable hard drives.

6.2 TPS Nodes and Domain Controller Considerations

Domain Controller usage with TPS

Microsoft provides a promotion tool (dcpromo), which can be used to upgrade a Windows 2003 Server to a Domain Controller.

In general, Honeywell does not recommend mixing applications with administration functions in the same hardware. An APP running in a small system can be setup to run inside a workgroup, rather than a domain, making a Windows domain unnecessary. For Windows 2000 and greater, the APP Solution Package works in a workgroup. In the Windows NT 4.0 version, a windows domain was required.

CAUTION

If you are implementing a TPS Domain OU, do NOT use a TPS APP node as the Domain Controller. Microsoft advises not to move a domain controller from the domain controller OU into a TPS domain OU. DO NOT put a Domain Controller inside a TPS Domain OU.



ATTENTION

Windows 2000 Domain Controllers (including the WINS server) must not be Fault Tolerant Ethernet (FTE) nodes.

Do not use a GUS node as a Domain Controller



ATTENTION

A GUS node must never be configured as the Domain Controller because the Windows 2000 Server operating system assigns display functions as the lowest priority.

6.3 System Operation During Domain Controller Outages

Although Windows 2003 Domain Controllers act as peers, if you have a need to run in mixed mode, the concept of a PDC and BDC still exists. Honeywell recommends that you DO NOT run in mixed mode unless absolutely necessary.

In mixed mode, the first Domain Controller in a Windows 2003 domain automatically receives the roles of PDC Emulator and RID Master. These roles, otherwise known as FSMO, allow a single Domain Controller to act in the role of a Windows NT 4.0 PDC.



REFERENCE - EXTERNAL

[Refer to Microsoft documentation for more information.](#)

6.4 Loss of a Domain Controller

Overview

While Domain Controllers play an integral role in the integrity of the network, loss of a Domain Controller is not a catastrophic network failure, especially when one or more peer Domain Controllers are available. This section lists the operations that can and cannot be performed, given a Domain Controller failure with a peer Domain Controller online, and a failure of all Domain Controllers.

What is supported if Root DC is offline and at least one other DC is online?

The following operations are supported when at least one Domain Controller is available and the Root Domain Controller is offline:

- Client applications can still make new connections to servers
- Previously configured domain nodes can start-up or shut down.
 - These nodes are properly represented in TPS Status Display
- Previously configured network users can log on to the nodes
 - These users will be logged on and will be able to access all TPS Components with the correct security access, like during normal operating conditions
- Application Module programs, and points will continue to operate as before
 - The APP Base provides AM functionality

What is NOT supported if Root DC is offline and at least one other DC is online?

The following operations are not supported when at least one Domain Controller is available and the Root Domain Controller is offline:

- Any network configuration change. (Mixed Mode only)
 - Adding new users or TPS components (such as data servers) to the domain is not possible.

What is supported if ALL Domain Controllers are offline?

The following operations are supported when all Domain Controllers are offline:

- Users currently logged on will continue to have access.
 - Users will already have acquired a security token from the domain controller; therefore, the client has security credentials to access other resources on the network.
 - Existing users that are currently logged on and want to continue to access the network must not log off.
- Previously configured nodes not already started can still start-up in local mode.
 - Only logging on locally is allowed. All components that are configured for AutoStart will not start automatically.

6 Recommendations for Managing a TPS System

6.4 Loss of a Domain Controller

- Displays currently in use and applications already connected to their data servers will continue to receive data.
 - A freshly invoked display will not be able to access HCI/OPC data (for example, data from OPC data sources). A freshly invoked display will get HOPC data (HOPC is an internal OPC-like standard and the data it provides is from the TPN) and PHD data using Honeywell ActiveX controls, but not HCI data (OPC data). Uniformance Desktop can still connect and get PHD data.
- Application Module programs and points would continue to operate.
 - The APP Base provides AM functionality and does not use the Domain Controller functionality.

What is NOT supported if ALL Domain Controllers are offline?

The following operations are not supported when all domain controllers are offline:

- New connections cannot be established between client applications and data servers.
 - New HCI-based displays cannot get data. New HOPC-based displays will work, provided the user account that is logged on has the appropriate permissions to run them. Note that in some cases this may mean the local account and not the domain account.
 - Fieldbus users cannot use the TPS Builder to connect to the TPN Server. TPS Builder will not be able to perform Fieldbus engineering functions.
- Users cannot log on to a node and use the domain account.
 - Users can log on to a node using a local account. For example, the local administrator account is accessible; but local logon accounts cannot run functions that require DCOM (Distributed Component Object Model), including the System Management Display and HCI-based GUS displays.
- Nodes started up in local mode would not be able to talk to other nodes.
 - These nodes would not be represented in the Honeywell System Management Display
- Applying security ACLs to resources is not possible.
 - The security configuration set up for the server is not available.

6.5 DLL Version Files

Overview

Prior to Windows 2000, DLL version file handling of DLLs presented problems. Problems occurred when each software application loaded its own set of specific system DLLs. Often, the newer DLLs were incompatible with other applications on the node. In some cases, the DLLs applied by various applications rendered the entire operating system unstable. The TPS system combated the problem by carefully screening which applications were qualified to run on a TPS node. In addition, a DLL version tool was provided to establish a DLL baseline list, which could be referred to in the event of a problem.

In Windows 2000 and Windows XP, Microsoft provides a technology that helps solve the DLL version problem. The Windows XP technology, known as Windows File Protection (Note: In windows 2000 it is called as System File Protection or SFP), allows key system files to be locked down. Individual applications are no longer able to modify these critical files. Attempts to do so result in the file automatically being replaced by the original. As a result, the need for a DLL version tool is reduced. The tool still plays a role, however, because the Microsoft Windows File Protection technology does not extend to vendor-specific DLLs, which can also cause serious problems.

DLL version file creation

After all software is loaded on each node, the DLL version tool can be used as described in the following procedure to create a baseline DLL version file. This file can be used later to compare the current DLL versions against the baseline DLL versions, as explained in the "DLL version file usage" section that follows.

Step	Action
1	Create the following directory: \\Program Files\\Honeywell\\DLLBaseline
2	Copy the following file: \\Program Files\\Honeywell\\tps\\baselldlldump.exe
3	Paste the file into the above "DLLBaseline" directory.
4	Select the following from the Start Menu: Programs > Accessories > Command Prompt

6 Recommendations for Managing a TPS System

6.5 DLL Version Files

Step	Action
5	Use the “cd” command to change to the following directory: \\Program Files\\Honeywell\\DLLBaseline
6	Issue the following command to create the DLL baseline file: <code>dlldump.exe > baseline.txt</code>  ATTENTION It takes several minutes to create the baseline file, which, typically, is over 200 KB.
7	Use the “exit” command to exit the Command Prompt window.
8	Copy the baseline.txt file to the “DLLBaseline” directory for safekeeping.
9	Name the copy, initialbaseline.txt .

DLL version file updates

If it is desired to create a new baseline file, then use the above creation instructions to write over the current “baseline.txt” file. Note that a copy of the initial baseline will still be available for future reference (initialbaseline.txt).

DLL version file usage

If it is suspected that a DLL file has been replaced with an incompatible version, then the following command can be used to compare the current DLL versions against the baseline DLL versions:

dlldump.exe baseline.txt



ATTENTION

The comparison will NOT work for DLL files located on FAT drives. This occurs because the tool performs an alphabetical comparison between the present DLL versions and the baseline DLL versions; and the DLL files on FAT drives are not listed alphabetically in the baseline DLL version file.

6.6 Allowable Desktop Sizes and Display Palettes

Resolution, color palette, and refresh rate recommendations on a GUS

A color palette of at least 65,536 colors (High Color 16 bit) is recommended to ensure that color stealing does not occur between applications.

The following table lists the colors and resolutions supported for GUS workstations.

CRT Configuration	Minimum Required Color Bit	Recommended Resolution	Optional Resolution (See Note 1)
Single CRT	High Color (16-bit)	1024 x 768 @ 75 Hz	1280 x 1024 @ 75 Hz
Over/Under CRTs	High Color (16-bit)	1024 x 1536 @ 75 Hz	1280 x 2048 @ 75 Hz
Side By Side CRTs	High Color (16-bit)	2048 x 768 @ 75 Hz	2560 x 1024 @ 75 Hz
FPD (Flat Panel Display) – Native Resolution	High Color (16-bit)	1280 x 1024 @ 60 Hz	1600 x 1200 @ 60 Hz max.

Note 1 – At the required minimum 65536 colors (16-bit) color setting, running any of the optional display resolutions will require a 4 MB video card. DELL and Motorola Pentium PRO-based GUS platforms have 2 MB video cards. A 4-MB dual video card upgrade kit is available for the dual CRT GUS, and can also be used with a single CRT GUS. The model number for that upgrade kit is TP-QCRTVB-100.



ATTENTION

A refresh rate of 75 Hz or higher is recommended for all CRT resolutions.
A refresh rate of 60 Hz is recommended for all Flat Panel Displays.



REFERENCE - INTERNAL

For Experion nodes, see “Configuring the Computer’s Display” in the *Experion Software Installation and Upgrade Guide* for details on resolution, color palette, and refresh rate.

6.6 SafeView Recommended For Process Views

Use SafeView when view of process is important

SafeView is recommended for operator and supervisor environments in order to avoid loss of view and to manage the window space easily. Using SafeView requires the engineer or administrator to define where and how big each window or "ViewPort" should be. Prior to defining SafeView ViewPorts, the size of the desktop needs to be determined, as well as which applications will (or will not) be managed by SafeView.

SafeView can manage most applications. Those that it cannot manage should not be installed or used in an operator environment.



REFERENCE - INTERNAL

For a list of these applications, refer to the "Exclusions and Special Considerations" section in the *SafeView User's Guide*.

In addition, the size of any SafeView window must be large enough, so the application is not constrained to a size where it is unusable. In other words, if the window requires a certain width and height, it should not be allowed to be made smaller than that.



REFERENCE - INTERNAL

For additional SafeView guidelines, refer to the "Design Considerations" section of the *SafeView User's Guide*.

6.7 Recommended Settings for Event Viewer Log

Set up for periodic reviews

It is recommended that you periodically review past events with the standard Microsoft Event Viewer application and archive the log files as necessary.



REFERENCE - INTERNAL

For more information about setting up event filtering, see the "Configuring Event Filtering" section in the *System Management Configuration Guide*.



REFERENCE - EXTERNAL

See the Microsoft online documentation for further information.

Honeywell recommends the following Event Log settings:

Event Log	Settings
System	MAX Size: 1024 K Event Wrapping: Overwrite Events as Needed
Security	MAX Size: 512 K Event Wrapping: Do NOT Overwrite Events as Needed
Application	MAX Size: 1024 K Event Wrapping: Overwrite Events as Needed

6.8 PC Performance Monitoring

Windows XP monitoring tool

Windows XP includes a powerful and flexible monitoring tool called the Performance Monitor. The tool can monitor several system- or application-specific variables using a variety of formats, including charts (trends) and alerts.



REFERENCE - EXTERNAL

[Refer to the online Microsoft documentation for additional usage details.](#)

Performance counters to monitor

Administrators should set up charts and alerts to suit their needs. The following counters are of particular use:

Object	Counter
Processor	% Processor Time
Memory	Available Bytes
Memory	Committed Bytes
Network	%Network Utilization Note: This value is not available by default. The value appears when the Microsoft Windows XP/2003 Network Monitor driver is installed. This driver is installed through the network card configuration by selecting Install > Protocol > Network Monitor Agent .
TPN Server	All counters

Honeywell System Performance Server

The Honeywell System Performance Server (SPS) is an OPC Data Access server that provides an efficient mechanism for retrieving performance and configuration information. This information is internally maintained on a per-node basis using Microsoft's Windows Management Instrumentation (WMI). The availability of this data through OPC interfaces allows it to be easily integrated into operator displays and process applications in a manner consistent with process data access.

By representing WMI data as OPC data items, the DCS infrastructure is able to display, trend, and/or historize Windows performance information. While the Microsoft monitoring tool allows trending and analysis of PC performance information, it cannot match the capabilities available in the process control environment, and it does not allow integration of the PC performance data into the analysis, historization, and trending of process control data.



REFERENCE - INTERNAL

The System Performance Server is an HCI Managed Component. For detailed information, refer to the "Role of the System Performance Server" section in the *System Management Configuration Guide*.

6.9 Saving Dr. Watson Dump Files

Default Dr. Watson dump file and its location

Dr. Watson dump files are useful for troubleshooting at the site, and also for calls to TAC (Technical Assistance Center). The following is the default Dr. Watson dump file and directory:

<Drive>:\WINDOWS\user.dmp

The following user groups do not have permission to write to the "WINDOWS" directory:

- Engineer
- Operator
- Supervisor

If a crash occurs while a user is logged on to one of the above groups, the user will be presented with a Dr. Watson prompt to browse to a new dump directory. The user should save the dump file for troubleshooting to the directory of their choice.

6 Recommendations for Managing a TPS System

6.9 Saving Dr. Watson Dump Files

7. Administrative Installation

7.1 Overview

A server-based administration solution is available for TPS software installation/upgrades, which can be published to TPS nodes. This mechanism allows you to publish a program distribution to users on the network.

After the user logs on to a computer that is part of a Windows domain for TPS nodes, a published program displays in the Add/Remove Programs dialog box. The program can be installed from there.

This feature is in addition to the existing option of TPS package installation on a local computer by the user.

7.2 Procedures to Support Network Install of TPS Software

Installation Tasks

To install the TPS software, perform the following tasks:

1. Create a Distribution Point in the server machine.
2. Perform Administrative Install (preparation) of GUS/APP common component software from the **TPS System Software CD**.
3. Perform Administrative Install (preparation) of GUS/APP TPS software from the **TPS System Software CD**.
4. Publish the GUS/APP software using Group Policy Objects.

Creating a Distribution Point

To begin, you need to create a Distribution Point and a directory structure for the TPS software packages to be installed. The Distribution Point should be accessible on the network as it acts as a host for the TPS packages/components to be installed.



ATTENTION

GUS and APP nodes require US/AM personality packages and TPS security packages to function properly. These software packages should be copied from the appropriate media to the Distribution Point before they can be published for network installation. This section provides instructions.

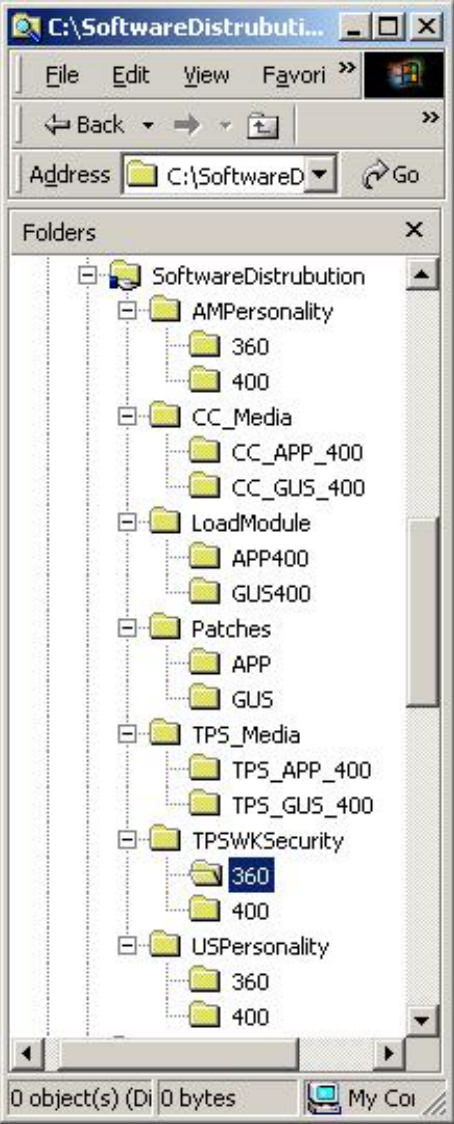
7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

To create a Distribution Point

You need to create an individual directory for each package that has an `install.exe` and `setup.exe` files. Follow these steps to create a Distribution Point in the Server machine.

Step	Action
1	Log on to a Server machine as a Domain Administrator. For example: <machine name>\<domain administrator> Note: The machine name example used in these procedures is <i>uis-msi-svr</i> .
2	Open Windows Explorer .
3	Select drive C .
4	Click File>New>Folder .
5	Name it as Software Distribution .
6	Double click the folder Software Distribution .
7	Click File>New>Folder .
8	Name the Folder as CC_Media to save the Common Components.
9	Double click the folder CC_Media .
10	Click File>New>Folder .
11	For GUS, name the folder as CC_GUS_400 . For APP, name the folder as CC_APP_400 .

Step	Action
12	Repeat Steps 3-11 to create folders for TPS Software, Personality, and Security. Name the folders as shown in the following example of a multiple directory structure. 

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7.2 Procedures to Support Network Install of TPS Software

Step	Action
13	Right -click Software Distribution folder.
14	Select Sharing... Software Distribution Properties page is displayed.
15	Click Share this folder option.
16	Click Permissions .
17	Click Add... and then select Administrator .
18	Click OK .
19	Highlight Administrator and then check Full Control .
20	On the Permissions for Software Distribution page, highlight Everyone and then click Remove .
21	Click Add and then select Authenticated Users .
22	Click OK .
23	Select Authenticated Users and then select the corresponding check box displayed against Read under Allow. Ensure the permission for Full Control and Change are cleared.
24	Click Apply and then click OK .
25	Select Security tab on the Software Distribution Properties page.
26	Select Everyone and then click Remove .
27	Click Advanced , and then clear Inherit from parent the permission entries that apply to child objects .
28	Click Remove on the Security page.
29	Click Add on Software Distribution Properties page and then select Authenticated Users .
30	Click Add and then click OK .

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

Step	Action
31	Select Authenticated Users and then select the corresponding check box displayed against Read & Execute , List Folder Contents and Read under Allow. Ensure the permission for Full Control, Modify, and Write are cleared.
32	Click Apply and then click OK .

After you complete the creation of the Distribution Point, proceed with the Administrative Install procedure that follows.

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

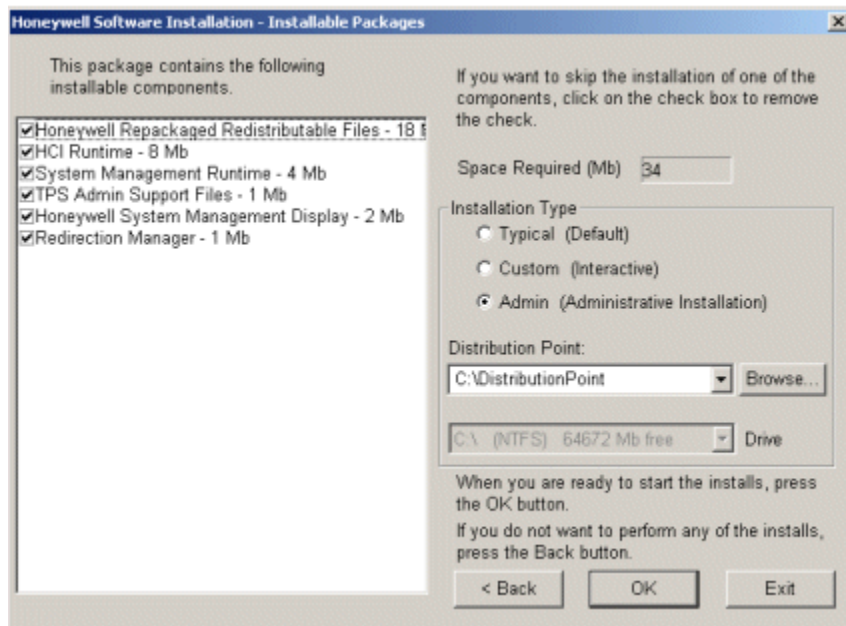
Administrative Install



ATTENTION

The Administrative Install does not install a package, but rather prepares your package to allow it to be installed from a network accessible distribution point

Click **Admin (Administrative Installation)** and then enter the path for a Distribution Point or Browse as shown in the following example:



The Admin Install interaction is automatically recorded in an `install.ini` file. This file is located on the Distribution Point where the `install.exe` is located. The `install.ini` is used for subsequent client installation, so that users on the client machine need not interact during the install. This is applicable to the TPS System Software CD.



ATTENTION

The client installation must be performed for each computer that needs to run the software application.

Creating a .zap file

The Publish mechanism requires a .zap file to be created.

A .zap file is a text file (similar to .ini files) that provides information about how to install a program, the application properties, and the entry points that the application should install. The .zap files are hosted on the Distribution point.

You can make use of the sample .zap files or generate a specific .zap file that appropriately describes the packages to be installed.



REFERENCE - EXTERNAL

Refer to HOW TO: Publish non-MSI Programs with .zap Files
<http://support.microsoft.com/default.aspx?scid=kb;en-us;231747>

Administrative Install of common components from TPS System Software CD

Follow these steps to install the Common Components for GUS/APP.

Step	Action
1	Log on to a Server machine as a Domain Administrator .
2	Insert the TPS System Software CD into the CD-ROM drive.
3	In Windows Explorer, browse to the Packages folder on the CD, and double-click the install.exe file.
4	Click Next on the Welcome page.
5	Click Next on the End User License Agreement page.
6	Click Next on the Third-Party Software Compatibility Policy page.
7	Enter the License No. and Authorization No. on the User Information page and then click Next .

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

Step	Action
8	For GUS, select the following packages: • Honeywell System Management Display • Redirection Manager For APP, select the following package: • Honeywell System Management Display
9	Click Install Package in the Package Selection page.
10	Click Admin Install .
11	Browse or enter the path to the Distribution point. For example, enter the following for GUS: C:\SoftwareDistribution\CC_Media\CC_GUS_400 For example, enter the following for APP: C:\SoftwareDistribution\CC_Media\CC_APP_400 Note: Even if you use the option from Start>Run..., you must click Admin at this point, if you want to do Admin Install. When you select Admin Install, the Silent Install is selected automatically.
12	Click OK .

After you have completed the installation of the common components, proceed with the procedure for the Administrative Install of the TPS software.

Administrative Install of TPS software from TPS System Software CD

Follow these steps to perform the Administrative install of the TPS Software for GUS/APP:

Step	Action
1	Log on to a Server machine as a Domain Administrator .
2	Insert TPS System Software CD into the CD-ROM drive.
3	In Windows Explorer, browse to the Packages folder on the CD, and double-click the <code>install.exe</code> file.
4	Click Next in the Welcome page.
5	Click Next in the End User License Agreement page.
6	Click Next in the Third-Party Software Compatibility Policy page.
7	Enter the License No. and Authorization No. in the User Information page and then click Next.
8	For GUS, select the following packages: • GUS Base Software • GUS Display Builder • GUS Display HCI Client Add-in • GUS Display Runtime • GUS Multiple Displays For APP, select the following packages: • APP Base Software • APP Solution Package • CL/NT Application Server
9	Click Install Package in the Package Selection page.
10	Click Admin Install .

7 Administrative Installation

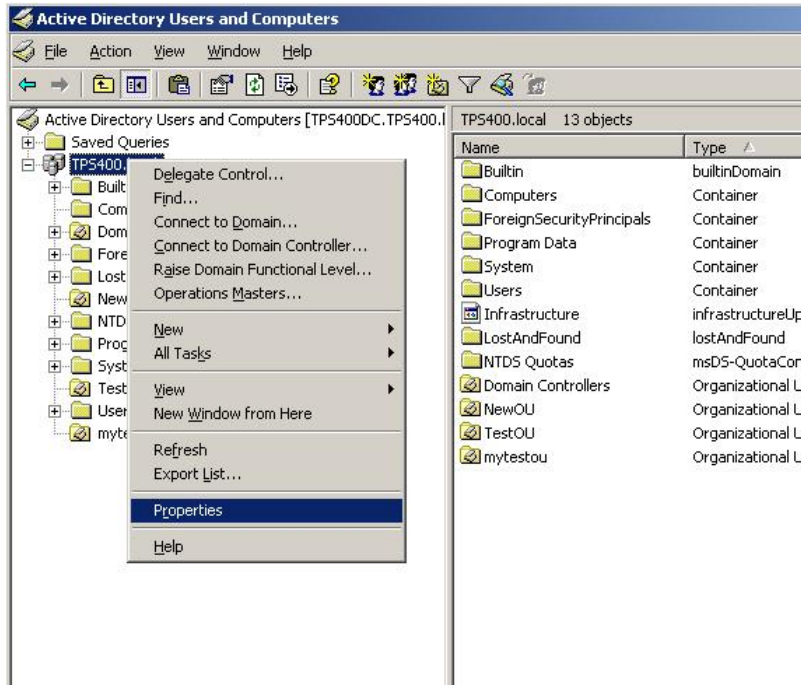
7.2 Procedures to Support Network Install of TPS Software

Step	Action
11	Browse or enter the path to the Distribution point. For example, enter the following for GUS: C:\SoftwareDistribution\TPS_Media\TPS_GUS_400. For example, enter the following for APP: C:\SoftwareDistribution\TPS_Media\TPS_APP_400. Note: Even if you use the option from Start>Run..., you must click Admin at this point, if you want to do Admin Install. When you select Admin Install, the Silent Install is selected automatically.
12	Click OK .

After you complete the Administrative Install of the TPS software, proceed with the procedure to publish GUS/APP software using Group Policy Objects.

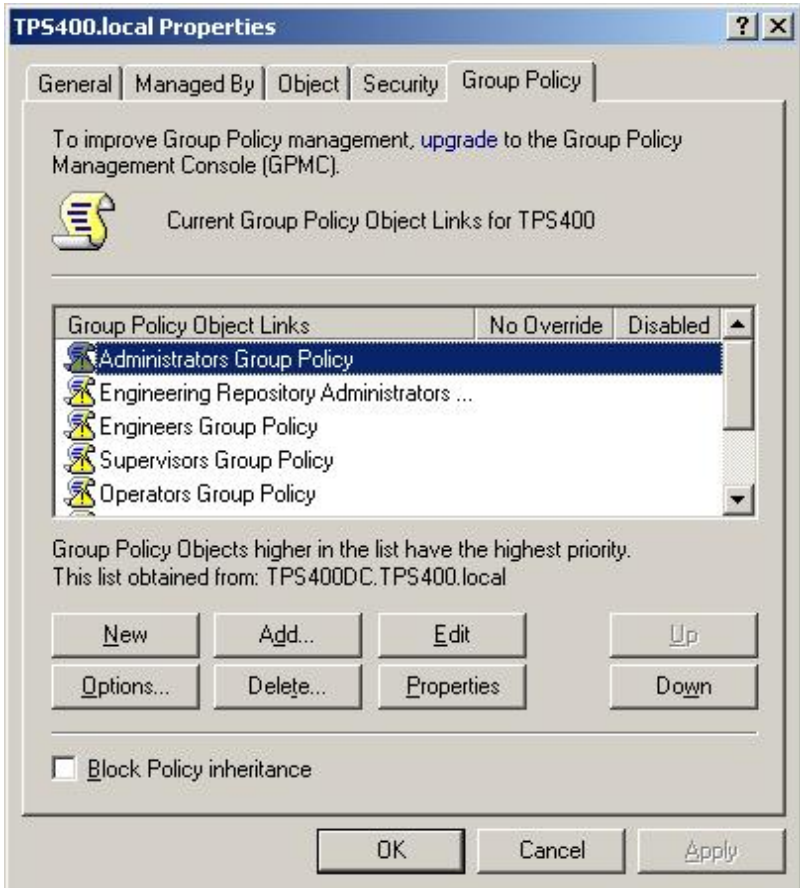
Publishing GUS/APP software using Group Policy Objects

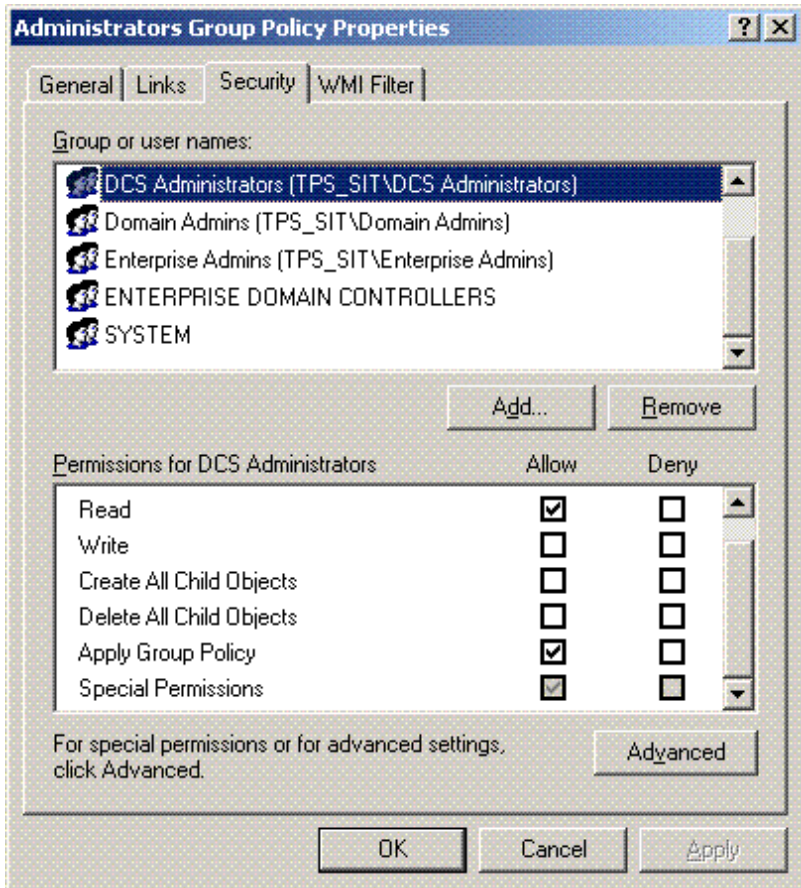
Perform the following steps to publish the GUS/APP software using the Group Policy objects:

Step	Action
1	Log on to a Server machine as a Domain Administrator .
2	Click Start> All Programs> Administrative Tools> Active Directory Users And Computers .
3	Right-click the desired Console and then select Properties. Note: For this example, the name of the Console is TPS400.local.  The screenshot shows the 'Active Directory Users and Computers' window for the domain 'TPS400.local'. The left pane shows a tree view with 'TPS400.local' selected. A right-click context menu is open over 'TPS400.local', and the 'Properties' option is highlighted. The right pane shows a list of objects in the domain, including 'BuiltIn', 'Computers', 'ForeignSecurityPrincipals', 'Program Data', 'System', 'Users', 'Infrastructure', 'LostAndFound', 'NTDS Quotas', 'Domain Controllers', 'NewOU', 'TestOU', and 'mytestou'.

7 Administrative Installation

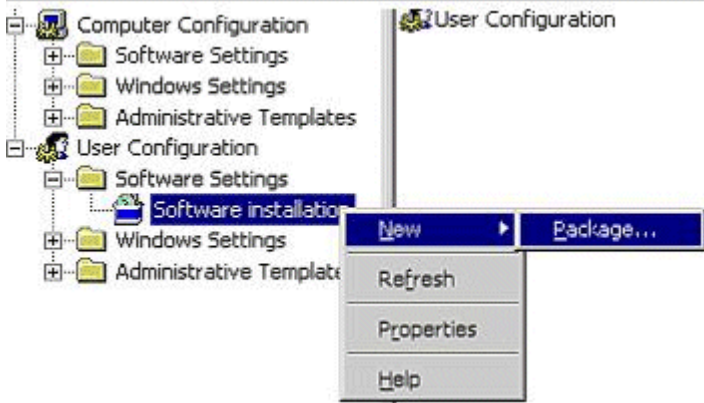
7.2 Procedures to Support Network Install of TPS Software

Step	Action
4	Click the Group Policy tab.  The screenshot shows the 'TPS400.local Properties' dialog box with the 'Group Policy' tab selected. The dialog contains a message about upgrading to the Group Policy Management Console (GPMC). Below this is a section titled 'Current Group Policy Object Links for TPS400' which contains a list of Group Policy Object Links. The list includes 'Administrators Group Policy', 'Engineering Repository Administrators ...', 'Engineers Group Policy', 'Supervisors Group Policy', and 'Operators Group Policy'. The 'Administrators Group Policy' is selected. Below the list are buttons for 'New', 'Add...', 'Edit', 'Up', 'Options...', 'Delete...', 'Properties', and 'Down'. At the bottom of the dialog is a checkbox for 'Block Policy inheritance' and 'OK', 'Cancel', and 'Apply' buttons.
5	Select Administrator Group Policy, and then click Properties.

Step	Action
6	Click the Security tab. Ensure the DCS Administrators group has only Read and Apply Group Policy permission as shown below.  The screenshot shows the 'Administrators Group Policy Properties' dialog box with the 'Security' tab selected. The 'Group or user names' list contains five entries: 'DCS Administrators (TPS_SIT\DCS Administrators)', 'Domain Admins (TPS_SIT\Domain Admins)', 'Enterprise Admins (TPS_SIT\Enterprise Admins)', 'ENTERPRISE DOMAIN CONTROLLERS', and 'SYSTEM'. The 'DCS Administrators' group is selected. Below the list are 'Add...' and 'Remove' buttons. The 'Permissions for DCS Administrators' table has two columns: 'Allow' and 'Deny'. The permissions listed are 'Read', 'Write', 'Create All Child Objects', 'Delete All Child Objects', 'Apply Group Policy', and 'Special Permissions'. The 'Read' and 'Apply Group Policy' permissions are checked in the 'Allow' column. The 'Advanced' button is located at the bottom right of the dialog box.
7	Click OK. Note: Do not use the Deny option, because it can cause unwanted results.
8	Click Edit on the TPS400.local Properties page.

7 Administrative Installation

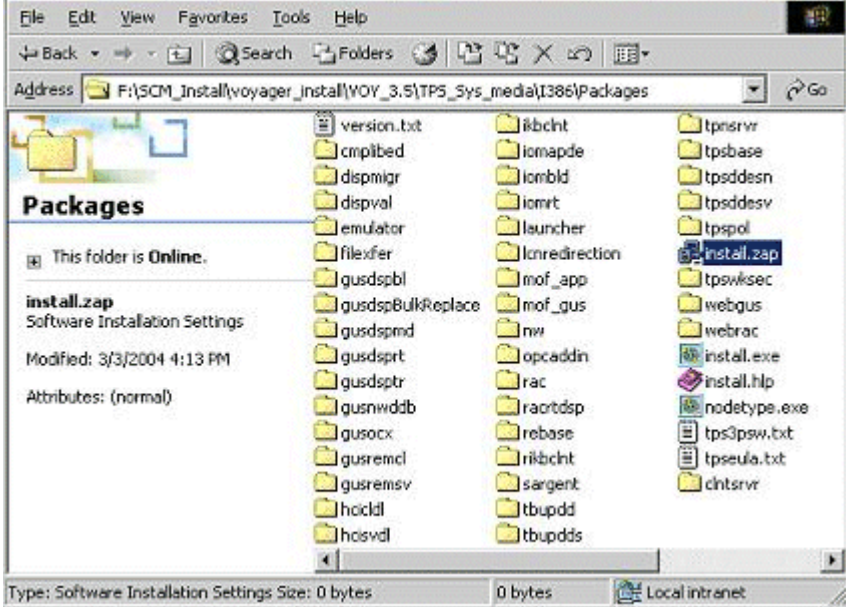
7.2 Procedures to Support Network Install of TPS Software

Step	Action
9	Browse to User Configuration\ Software Settings\ Software installation.
10	On Group Policy page, right-click Software installation, and then select New>Package... 
11	For GUS, select: My Network Places> Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr> SoftwareDistribution> TPS Media> TPS_GUS_400 For APP, select: My Network Places> Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr> SoftwareDistribution> TPS Media> TPS_APP_400
	ATTENTION Verify that the cursor is pointing to the Software Distribution Point corresponding to the network and not to the local drive on the Server.

Step	Action
12	Use Notepad to create the <code>install.zap</code> file shown below. For GUS: <pre>[Application] FriendlyName = "CC GUS - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\CC_Media\CC_GUS_400\install.exe" DisplayVersion = 400;</pre> For APP: <pre>[Application] FriendlyName = "CC APP - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\CC_Media\CC_APP_400\install.exe" DisplayVersion = 400;</pre> Note: The text in quotes (after <code>SetupCommand</code>) needs to be on a single line.
13	For GUS: Save the <code>install.zap</code> file in SoftwareDistribution\ CC_Media\ CC_GUS_400. For APP: Save the <code>install.zap</code> file in SoftwareDistribution\ CC_Media\ CC_APP_400.

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

Step	Action
14	Select Files of type as *.zap, and then select <code>install.zap</code>. 
15	Click Open .
16	In the Deploy Software dialog box, click Publish , and then click OK .
17	In the Group Policy page, right click Software installation , and then select New>Package...

Step	Action
18	For GUS, select: My Network Places >Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr > SoftwareDistribution> TPS_Media> TPS_GUS_400 For APP, select: My Network Places> Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr > SoftwareDistribution > TPS_Media> TPS_APP_400
	ATTENTION Verify that the cursor selects the Software Distribution Point that corresponds to the network drive and not to the local drive on the Server.
19	Create install.zap file shown below using notepad. For GUS: <pre>[Application] FriendlyName = "TPS GUS - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\TPS_Media\TPS_GUS_400\install.exe" DisplayVersion = 400;</pre> For APP: <pre>[Application] FriendlyName = "TPS APP - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\TPS_Media\TPS_APP_400\install.exe" DisplayVersion = 400;</pre> Note: The text in quotes (after SetupCommand) needs to be on a single line.

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

Step	Action
20	For GUS: Save the <code>install.zap</code> file in SoftwareDistribution\ TPS_Media\ TPS_GUS_400. For APP: Save the <code>install.zap</code> file in SoftwareDistribution\ TPS _Media\ TPS APP_400.
21	Select Files of type as <code>*.zap</code> , and then select <code>install.zap</code> .
22	Click Open .
23	In the Deploy Software dialog box, click Publish , and then click OK .
24	For GUS: From GUS TPN Software CD, copy GUS\PERS_US folder into SoftwareDistribution\ USPersonality\ 400 folder For APP: From TPN Application Module CD, copy APP\PERS_AM folder into SoftwareDistribution\ AMPersonality\ 400 folder.
25	In the Group Policy page, right click Software installation for TPN Personality Software, and then select New>Package....
26	For GUS: Select My Network Places> Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr> SoftwareDistribution> USPersonality> 400> PERS_US> disk1. For APP: Select My Network Places> Entire Network> Microsoft Windows Network> TPS400.local> uis-msi-svr> SoftwareDistribution> AMPersonality> 400>PERS_AM> disk1.

Step	Action
27	Use Notepad to create the setup . zap file shown below. For GUS: <pre>[Application] FriendlyName = "GUS Personality - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\USPersonality\400\setup.exe" -uSTPS:D_C:T_T:ETPS DisplayVersion = 400;</pre> For APP: <pre>[Application] FriendlyName = "APP Personality - 400" SetupCommand = "\\uis-msi-svr\Software Distribution\AMPersonality\400\setup.exe" -uSTPS:D_C:T_T:ETPS DisplayVersion = 400;</pre> Note: The text in quotes and the command line arguments (after SetupCommand) needs to be on a single line.
28	For GUS: Save the setup . zap file in SoftwareDistribution\ USPersonality\ 400\ PERS_US\ disk1 directory . For APP: Save the setup . zap file in SoftwareDistribution\ AMPersonality\ 400\ PERS_AM\ disk1 directory.
29	Select Files of type as *.zap, and then select setup . zap .

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

Step	Action
30	Click Open .
31	In the Deploy Software dialog box, click Publish , and then click OK .
32	From the TPS System Software CD, copy D:\I386\PACKAGES\wksecurity folder into SoftwareDistribution\TPSWKSecurity\400 folder.
33	In the Group Policy page, right-click software installation to install the High Security Policy Package and then select New>Package .
34	Select My network Places> Entire network> Microsoft windows network> TPS400.local> uis-msi-svr > SoftwareDistribution> TPSWKSecurity> 400> tpswksec .
35	Use Notepad to create a setup.zap file: [Application] FriendlyName = "Workstation Security - 400" SetupCommand = setup.exe /unattend DisplayVersion = 400;
36	Put setup.zap file on SoftwareDistribution\ TPSWKSecurity\ 400\ tpswksec directory.
37	Select Files of type as *zap, then select setup.zap.
38	Click Open .
39	In the Deploy Software dialog box, select Publish and then click OK .



REFERENCE - INTERNAL

For the procedure to perform a Network Installation on a target client machine, refer to the *TPS System Implementation Guide*.

Converting .zip file to .exe file

The packages have to be in .exe format for network installation. Perform the following steps to convert .zip format to .exe format.

Step	Action
1	In the Windows Explorer browse to the .zip file and double-click. For this example, the .zip file is GUS_330.1.3.zip . The WinZip window appears.
2	Click Actions>Make.exe File . The screenshot shows the WinZip application window titled 'WinZip - GUS_330.1.3.zip'. The 'Actions' menu is open, displaying various options. The 'Make .Exe File' option, which has a keyboard shortcut of 'Shift+K', is highlighted in blue. Other visible options include 'Add...', 'Delete...', 'Extract...', 'View...', 'Select All', 'Invert Selection', 'Virus Scan', 'UUencode', 'Test', 'Comment...', 'CheckOut...', and 'Install...'. The background of the window shows a file list with columns for 'Name', 'Type', and 'Modified'.
3	Click OK in the WinZip Self-Extractor Personal Edition . The .zip file is converted to an .exe file.

7 Administrative Installation

7.2 Procedures to Support Network Install of TPS Software

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